

# SHIFTWM: OBSERVATION-ANCHORED SPATIAL MIXING FOR COMPACT WORLD MODELS

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## ABSTRACT

Visual world models must predict how a scene changes after an action while preserving useful information from real observations. We propose ShiftWM, a compact predictor that keeps the last observed visual feature grid as a fixed reference. For each future action prefix, it mixes features from this grid and adds a bounded correction. This gives every forecast a direct path to measured visual evidence. In a matched development study on recorded DROID robot video, ShiftWM reduces ten-step endpoint feature error by 5.3% relative to autoregression. Component experiments support action-conditioned mixing as the clearest contributor to the improvement. We also analyze this decoder under single-image forecasting on recorded pushing and bimanual manipulation. We provide trained checkpoints, reproducible evaluation code and visual diagnostics that show where the method improves predictions and where it still fails.

## 1 INTRODUCTION

Visual world models predict the consequences of actions from images. A common approach rolls a predictor forward, feeding each predicted visual state into the next step. This makes a long forecast depend increasingly on its own estimates. The last real observation, however, remains a measured description of the scene. We ask how a compact model can keep that evidence available while predicting action-dependent changes.

We propose **ShiftWM**, an observation-anchored spatial predictor built from released LeWorldModel components (Maes et al., 2026b). Its decoder keeps the last observed feature grid fixed. For each supplied action prefix, it predicts how to mix features from that grid and adds a bounded correction. The model can therefore change the predicted state without replacing its observed visual reference with an imagined one. Figure 1 contrasts these two routes to a forecast. Its input gallery also locates the recorded-video transfer tasks and simulation studies documented in the attached appendix.

We evaluate this design through a matched study on recorded DROID robot video. The proposed model reduces native ten-step endpoint feature error by **5.30%** relative to autoregression and **3.55%** relative to an additive observed-state anchor. The component experiments support the *action-conditioned mixing package*; they do not establish an incremental native-error advantage for bounding or the separate support-context path. We report the complete model and controlled removals under one matched evaluation protocol. Separately trained single-image adapters test the same decoder on recorded IWS pushing and bimanual manipulation tasks, under a distinct forecasting and selection protocol.

Our contributions are the fixed-observation spatial decoder, a matched component analysis, and reproducible real-video evaluations with checkpoint and visual evidence. These are development findings. Earlier context-model studies, including mixed planning and extrapolation outcomes, remain explicitly separate in the attached appendix and do not establish performance of the spatial model.

## 2 TASK AND AVAILABLE INFORMATION

Given an observed history and supplied actions, predict future visual features. The DROID history contains three real images and two past action blocks; the query contains ten future blocks. Each



Figure 1: **Forecast from retained observations.** Three recorded DROID frames supply fixed support features and the last-observed reference  $Z_0$ ; the palette compresses this interface (Figure 2). ShiftWM conditions every query on this support and supplied causal commands; autoregression rolls its own predictions into the next feature history, keeping inferred context fixed. Two schematic endpoints illustrate feature forecasts, not generated RGB; the camera is illustrative. Historical simulations use distinct models; IWS tests a separately trained single-observation adapter; Open-H is a physical-phantom input audit. All 141 DROID development episodes remain visible. The 5.30% reduction compares population mean native, training-standardized  $h = 10$  feature MSE with matched autoregression; uncertainty: Figure 3. Images: DROID/Open-H (CC BY 4.0); IWS (Zhang et al., 2026).

block groups five native recorded commands. A frozen encoder maps images into a spatial grid. Withheld future images supply evaluator targets and never enter support or context inference. The task measures feature forecasting, with no generated RGB or physical robot-control evaluation. Figure 2 shows the model-visible computation; Figure 4 shows recorded observations and measured forecast errors.

### 3 OBSERVATION-ANCHORED SPATIAL PREDICTION

ShiftWM predicts a future feature grid from fixed observed evidence and a causal action prefix. Figure 2 links the fixed observed memory, horizon-specific conditioning and patchwise decoder, with withheld target features kept in a separate scoring path.

#### 3.1 OBSERVED SUPPORT AND CAUSAL ACTION CONDITIONING

Let  $Z_0 \in \mathbb{R}^{16 \times 384}$  be the last observed DINOv2-small grid, standardized with training-only channel statistics. A shared spatial encoder maps each observed grid into 96-dimensional patch embeddings. In the DROID instantiation, a single support context summarizes only observed transitions. A separate unidirectional GRU encodes the chronological action prefix, and the reused temporal predictor produces a horizon-specific patch state  $H_h$ . The observed support stays fixed as the query horizon increases; imagined images never become new support evidence. Feature standardization is inverted when predictions are returned in the encoder’s coordinates. The single-image IWS adapter repeats the initial grid into three architectural slots and omits transition context and FiLM; it does not invent observed history.



Figure 2: **ShiftWM (ours): retain observed features, compose a new forecast.** (A) Recorded DROID frames 0/5/10 become three frozen-DINOv2 grids with fixed training-only channel normalization. The learned spatial encoder processes all three grids into  $E_{-2:0}$ ; its last pre-FiLM encoding  $E_0$  and source  $Z_0$  stay fixed throughout forecasting. Only DINOv2 weights are frozen. (B) Context from patch means, their chronological differences and past commands conditions the three encodings; a separate chronological action GRU and temporal predictor produce  $H_h$  from the supplied prefix. (C)  $Q(H_h)$  and  $K(E_0)$  determine a row-softmax matrix with the  $+4I$  bias outside  $\sqrt{96}$  scaling. The highlighted row  $i = 6$  mixes all sixteen observed patch vectors; one-based contributors 1/6/11 are illustrated and patch 6 supplies the direct branch in (D). (D) The same  $H_h$  sets the affine sigmoid gate and, independently, a LayerNorm–affine  $96 \rightarrow 384$  projection followed by  $\tanh$ . The complementary blend receives this unit-bounded update in standardized feature coordinates after summation; see Eqs. (1)–(2). The lower lane scores the assembled feature grid against the withheld frame through the same frozen encoder and normalization, with no target-to-predictor path. Feature pieces, weights and channel bars are schematic, not RGB forecasts. DROID photographs are unchanged (CC BY 4.0).

### 3.2 A FIXED ANCHOR, GATED MIXING AND BOUNDED CORRECTION

Let  $E_0$  denote the last observed spatial encoding *before* context modulation. Queries from  $H_h$  and keys from  $E_0$  give

$$T_h = \text{softmax}_{\text{row}} \left( \frac{Q(H_h)K(E_0)^\top}{\sqrt{96}} + 4I \right), \quad g_h = \sigma(G(H_h)). \quad (1)$$

The identity bias favors preserving patch identity initially. The affine gate head  $G$  acts on each patch and broadcasts over channels; its bias initializes at  $-3$ . The prediction is

$$\hat{Z}_h = (1 - g_h) \odot Z_0 + g_h \odot (T_h Z_0) + \tanh R(H_h). \quad (2)$$

The residual head  $R$  uses normalization and an affine projection. Both the mixture source  $Z_0$  and key representation  $E_0$  come from the last observation at every horizon. Mixing semantic feature patches need not correspond to physical pixel motion.

Writing  $D_h = \text{diag}(g_h)$  gives row-convex weights  $W_h = I - D_h + D_h T_h$ . Hence each output channel lies between  $\min_k Z_{0,k,c} - 1$  and  $\max_k Z_{0,k,c} + 1$  in standardized coordinates. This bounds output range, not forecast accuracy or stability. We test bounding’s incremental benefit by removal.

### 3.3 TRAINING AND COMPONENT CONTROLS

All DROID component arms use the same frozen features, training data and selection recipe. We compare autoregression and persistence with the proposed model, then remove mixing, bounding, actions or the separate support-context path. Removing both mixing and bounding gives an additive anchor; adding bounding back yields a bounded-anchor control. These are controlled alternatives, not separately proposed algorithms. The mixing package includes its gate and identity bias and, in DROID, adds about 1.84% active parameters; the comparison therefore tests that package rather than a capacity-identical isolated operation. Exact objectives, dimensions, temporal indexing and stable checkpoint names are in Appendix B.

### 3.4 RELATION TO EXISTING WORLD MODELS

We reuse pinned LeWM prediction components (Maes et al., 2026b;a); compactness is an upstream property. ContextWM and ICWM condition predictions on scene or interaction histories (Wu et al., 2023; Wang et al., 2026b). MoVie and AdaJEPa study visual or online adaptation (Yang et al., 2023; Wang et al., 2026c); our evaluated weights remain fixed at inference. ReDRAW and RLA-WM use residual modeling (Lanier et al., 2025; Zhang et al., 2026), so a residual head alone is not our contribution. We investigate the *fixed observed source* of spatial mixing and its controlled alternatives. DA-LeWM motivates separating prediction quality from planning-cost alignment (Wang et al., 2026a). Our comparisons are matched design controls, not independent reproductions establishing state-of-the-art superiority.

## 4 EVALUATION DESIGN

Appendix A indexes the benchmark-specific metrics, complete method scorecards and outstanding external comparisons.

**Data and matched information access.** The selected DROID collection contains 1,126 recordings, split by recording session into 851 training, 143 validation and 132 original test episodes. The spatial study uses only training and validation. At ten query steps, 141 validation episodes from 59 sessions yield 1,631 eligible windows. All DROID scores use the common training-standardized  $4 \times 4$  grid; pooled  $2 \times 2$  coordinates are supplementary. Compared predictors share features, history and command access except for each explicitly removed path. Persistence copies the last observation.

The principal reported endpoint is native-grid standardized feature MSE at query step ten. For a comparator with error  $e_b$  and ShiftWM error  $e_o$ , we report  $100(e_b - e_o)/e_b$  as relative error reduction; this is not a percentage-point change in task success. Five-step endpoints and the full forecast curves remain visible as secondary diagnostics.

**Training and uncertainty.** The initial campaign trains five arms with three seeds for 30 epochs. A registered follow-up adds six models to complete the mixing-by-bounding comparison after the initial controls were revealed. All 21 runs complete the full budget. Checkpoints minimize window-weighted validation loss across all ten query steps. Reported endpoint scores average windows within episodes, then weight episodes and seeds equally. Paired intervals resample sessions and training seeds with 10,000 draws. These are exploratory development comparisons without multiplicity correction; the ten-step endpoint is not an untouched confirmatory test. All horizons, 36 paired-bootstrap contrasts (32 method comparisons and four interaction effects), selected checkpoints and exact reload checks appear in Appendix B.

**Single-observation transfer.** The IWS study trains separate task adapters on recorded PushT, Box and Rope trajectories. Each receives one image and 60 native commands and predicts the next 59 feature grids, with no observed-transition context or FiLM. Three learned arms, three tasks and three seeds give 27 full-budget runs; persistence is untrained. Checkpoints minimize development endpoint MSE. The primary comparison is bounded mixing versus additive anchoring, with paired seed-trajectory intervals and an equal-task mean of relative gains. Appendix C gives the complete input, split and selection contracts. Reserved upstream-validation outcomes remain separate.

**Historical studies remain distinct.** The earlier two-context model has different supervision and representations. Its simulation, original real-video test and fresh-session confirmation are preserved in Appendices I, D and E. Their primary outcomes and mixed findings are unchanged. They are neither ablations of this spatial predictor nor evidence of its cross-domain or closed-loop performance.

## 5 RESULTS AND ABLATIONS

**Forecasting from the observed anchor.** ShiftWM reduces native five-step endpoint error by **3.01%** and ten-step endpoint error from 0.203950 for matched autoregression to 0.193141, a **5.30%** relative reduction, and improves over the additive-anchor control by **3.55%**. The ten-step contrasts against autoregression and additive anchoring have paired intervals excluding zero. Table 1 gives exact endpoint scores; Figure 3 puts temporal behavior and paired gains side by side. Pooled-coordinate gains are smaller: 2.54% versus autoregression and an inconclusive 0.47% versus the additive anchor.

| Method                                  | h5 MSE ↓      | h10 MSE ↓     |
|-----------------------------------------|---------------|---------------|
| Autoregressive                          | 0.149573      | 0.203950      |
| Persistence                             | 0.164028      | 0.226129      |
| <b>ShiftWM (ours)</b>                   | 0.145074      | 0.193141      |
| <i>Component ablations</i>              |               |               |
| No mixing (bounded anchor)              | 0.149347      | 0.200010      |
| No bounding                             | 0.145037      | 0.193274      |
| No mixing or bounding (additive anchor) | 0.149606      | 0.200244      |
| No support context                      | 0.145143      | 0.193330      |
| No actions                              | 0.149480      | 0.202249      |
| ShiftWM gain vs autoregression          | <b>+3.01%</b> | <b>+5.30%</b> |

Table 1: **Matched spatial forecasting on DROID development data.** Native  $4 \times 4$  standardized endpoint feature errors average windows within episodes, then episodes and three seeds equally. All 21 trained models completed 30 epochs; persistence has no trained predictor. The last row reports relative reductions for ShiftWM against autoregression. Green identifies positive point gains; paired uncertainty and component effects appear in Figure 3 and Appendix B.

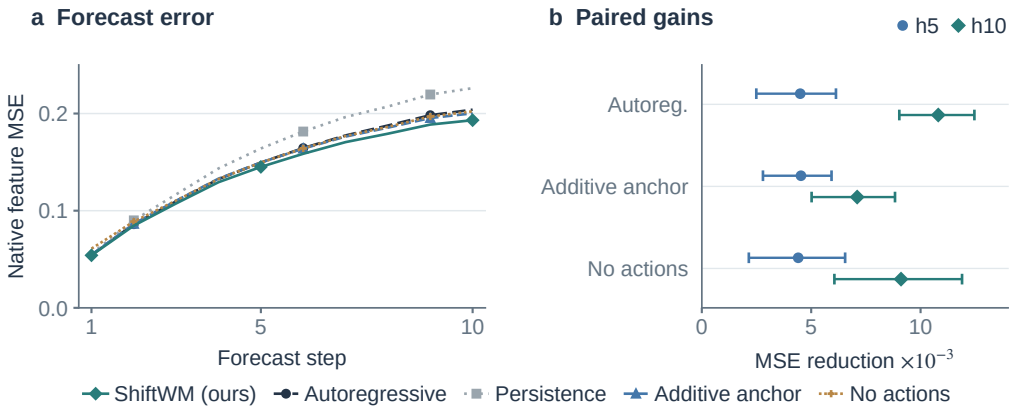


Figure 3: **Forecast accuracy and paired improvements.** (a) Native mean feature error at each step on DROID development data. (b) Comparator-minus-ShiftWM endpoint MSE at steps 5 and 10, with the original paired 95% session/seed bootstrap intervals. Both error and gain axes retain zero. The interval unit is MSE, not relative percent. Table 1 retains every spatial arm; Appendix B retains all 36 contrasts, including null and unfavorable ablations. Each forecast step spans five native commands.

**Ablations identify the clearest contributing package.** Adding mixing reduces native ten-step error by 3.48% without bounding and 3.43% with bounding; both paired intervals exclude zero. Incremental native bounding effects are inconclusive at both reported horizons, and every mixing-by-bounding interaction interval contains zero. The unbounded ablation has a slightly better five-step point estimate; the proposed bounded model has the lowest ten-step mean among these trained arms. Neither bounded nor unbounded mixing dominates across horizons. Two small pooled-coordinate bounding effects without mixing have favorable intervals. The complete component map and all null or unfavorable contrasts remain in Appendix B.

ShiftWM also reduces ten-step error by 4.50% relative to its action-free ablation. Removing the support-context path changes native ten-step error by only 0.10%, and all four context contrasts are inconclusive. Context removal also reduces active parameters by 14.40%. The evidence therefore supports the action-conditioned decoder package without establishing added benefit from the separate learned support context.

**What the recorded examples show.** Across all 141 validation episodes, 136 have lower native ten-step error under ShiftWM and five regress. Figure 4 shows the prespecified largest gain, median and largest regression on recorded scenes. Each pair uses the same withheld target and a common error scale; forecast-step diagnostics show when the methods separate. The appendix adds measured mixing weights and gates for these same cases. Per-episode records and the complete replay accompany the artifacts.



**Figure 4: Prespecified best, median and worst cases.** The episode-gain ranks among 141 DROID development recordings each show their first eligible window. AR denotes matched autoregression. Full RGB frames are recorded observations ( $f10$ ) and withheld targets ( $f60$ ). Maps show three-seed native, training-standardized  $h = 10$  feature MSE on one unclipped logarithmic scale. Curves show mean paired AR-minus-ShiftWM error with common  $\pm 70 \times 10^{-3}$  MSE limits; bands show seed minima/maxima, not confidence intervals. Teal gains mean lower error; amber shows regressions. Each forecast step consumes five native commands. Episode (Ep) gains average windows; the shown median window instead regresses. Images: DROID (CC BY 4.0).

**Single-observation decoder transfer.** A separate 27-model study tests one observed image and 60 native command rows on real IWS PushT, Box and Rope recordings. Table 2 reports all four predictors at stored offset 59 ( $H = 60$ ). The equal-task mean relative reduction against additive anchoring is +1.42% (paired 95% interval [+1.15, +1.67]%). Two of three tasks have a favorable point estimate against additive anchoring. Autoregression has lower endpoint error than ShiftWM on all three tasks. Thus the DROID improvement does not extend to a universal advantage for fixed-source decoding. These are separately trained development comparisons; complete curves, uncertainty and interfaces appear in Appendix C.

Table 2: Single-observation IWS development endpoints. Lower standardized feature MSE is better; the lowest mean per task is bold. Gains compare ShiftWM with the additive anchor; both point estimates and paired 95% intervals are in percent. Positive point estimates are bold green, whether or not their interval excludes zero. All tasks, matched seeds and controls are retained.

| Task  | Persistence | AR             | Additive anchor | ShiftWM (ours) | Gain vs anchor [95% CI]      |
|-------|-------------|----------------|-----------------|----------------|------------------------------|
| PushT | 0.57166     | <b>0.26000</b> | 0.26743         | 0.26112        | <b>+2.36%</b> [+2.09, +2.63] |
| Box   | 0.63039     | <b>0.27941</b> | 0.29656         | 0.29093        | <b>+1.90%</b> [+1.41, +2.39] |
| Rope  | 0.61714     | <b>0.21198</b> | 0.22289         | 0.22291        | -0.01% [-0.68, +0.59]        |

## 6 SCOPE AND REUSABLE ARTIFACTS

The results support a specific predictive design: retain measured visual features and learn action-conditioned mixing and correction. A fresh held-out spatial evaluation is still needed for a confirmatory superiority claim. The revealed original and fresh tests remain excluded from further model selection. The study does not establish physical control, zero-shot cross-embodiment generalization or an advantage over independently reproduced state-of-the-art models.

The spatial release contains all 15 initial predictors, including comparison arms, with source, normalization and checkpoint manifests. Six additional component predictors are available locally. All 21 pass exact relocated CPU reload checks. Earlier releases and other protocols remain in the appendix. The 27 selected IWS predictors also have local inference-only exports with exact relocated CPU reload checks; their feature-input interface and release scope are documented in Appendix C.

The central finding is that a compact predictor can benefit from continuing to use the last observed spatial state. Complete ablations, uncertainty and visual failures identify which parts of this design are supported and which need further testing.

## AI ASSISTANCE DISCLOSURE

AI assistance was used for drafting, public-source inspection, code development and editable figures. Human authors must verify the final text, sources, implementation and results and remain responsible for the submission.

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## GUIDE TO THE ATTACHED APPENDIX

The appendix continues this manuscript. Its order follows the main paper’s questions, then provides the earlier protocols and complete run records.

| Evidence group              | Where to look                                                                                                    |
|-----------------------------|------------------------------------------------------------------------------------------------------------------|
| Current method and transfer | Appendices B and C: exact decoder, DROID ablations, all 36 contrasts, and complete IWS forecast comparisons.     |
| Recorded-video history      | Appendices D–H: original test, fresh-session confirmation, and separately labeled development studies.           |
| Historical context model    | Appendix I: full architecture, paired losses, composition design, forecast and primary planning comparisons.     |
| Protocol and diagnostics    | Appendices J onward: exact simulation settings, information access, statistics, revisions and domain extensions. |
| Reusable artifacts          | Final appendices: model availability, visual provenance, training records and individual evaluation ledgers.     |

## A BENCHMARK COVERAGE AND COMPARISON COMPLETION

We organize comparisons by task, prediction interface and evaluation population. Tables 3 and 4 separate measured comparisons from external methods that still require matched evaluation. A training seed or checkpoint is not a distinct method. The current spatial decoder and the historical context model remain separate studies.

| Benchmark    | Task                         | Reported metrics              | Completed comparison / location                       |
|--------------|------------------------------|-------------------------------|-------------------------------------------------------|
| DROID        | Recorded robot forecasting   | Native / pooled MSE; h5 / h10 | 8 predictors; development; Table 5                    |
| IWS PushT    | Recorded pushing forecast    | MSE / MAE / L1 / cosine       | 5 predictors; development; Table 14                   |
| IWS Box      | Recorded bimanual forecast   | MSE / MAE / L1 / cosine       | 5 predictors; development; Table 14                   |
| IWS Rope     | Recorded deformable forecast | MSE / MAE / L1 / cosine       | 5 predictors; development; Table 14                   |
| Sim. PushT   | Closed-loop goal reaching    | Success; MSE@1/3/5            | 6 predictors; original test; Table 39                 |
| Sim. Reacher | Closed-loop goal reaching    | Success; MSE@1/3/5            | 6 predictors; original test; Table 39                 |
| Sim. drone   | Planar flight goal reaching  | Success; MSE@1/3/5            | 6 configurations + 2 revisions; development; Table 45 |
| Sim. tissue  | Tissue-point goal reaching   | Success; MSE@1/3/5            | 6 configurations; development; Table 45               |
| Open-H       | Physical-phantom input audit | No performance metric         | 0 evaluated methods; acquisition only                 |

Table 3: **Benchmark and result-table index.** Counts include ours and internal ablations, but exclude repeated seeds. DROID and IWS use the current spatial model; the simulations use the historical context model. Drone/tissue have three method families on two backbones; their final model tests remain pending. The separately registered IWS unbounded component study is counted only after its complete-campaign gate passes. MSE/MAE/L1/cosine are feature errors (lower is better); success is higher is better. Open-H is an input illustration, not an evaluated forecasting benchmark.

**Reading the numerical scorecards.** Each consolidated scorecard names the method and backbone, uses a fixed population within each comparison group, and identifies the metric direction and horizon. Black bold marks the lowest error or highest success point mean within that group, including a baseline when it wins. Green bold marks a positive gain against the explicitly named reference; neither style establishes statistical significance. Paired intervals remain in the accompanying inference tables. The proposed method is labeled ShiftWM (ours); removing a component defines an

ablation, not a separately proposed algorithm. An exploratory development winner does not silently replace the selected model or its earlier reported results.

| Benchmark              | External method | Target metrics                    | Execution status / required match                            |
|------------------------|-----------------|-----------------------------------|--------------------------------------------------------------|
| PushT / Reacher        | Frozen LeWM     | Success; forecast MSE             | Completed in historical grid; separate upstream check        |
| PushT                  | AdaJEPA         | Success under matched budget      | Official reproduction complete; matched comparison pending   |
| PushT / Reacher        | DINO-WM; PLDM   | Success; forecast MSE             | Not run; match visual inputs and planner budget              |
| Drone                  | PPO; PID        | Success; physical error; failures | Not run; disclose privileged state inputs if used            |
| Tissue                 | LapGym PPO      | Success; goal distance; failures  | Not run; match task and observation contract                 |
| DROID                  | LeWM; DINO-WM   | Feature MSE by horizon            | Not run under current forecast contract                      |
| IWS PushT / Box / Rope | RLA-WM          | Common-coordinate feature error   | Not run; resolve backbone access and representation mismatch |

Table 4: **External-comparison completion plan for this research draft.** Pending comparisons are not claimed as completed baselines. Internal controls and component ablations remain in the numerical tables. AdaJEPA uses different inputs and planning settings in its completed reproduction. RLA-WM uses a different feature representation; native scores are not interchangeable with ours. There is currently no completed matched external-method comparison for the spatial recorded-video model. Pending cells are unavailable, not zero performance.

**Additional measurements and completion rules.** Physical goal error and actions to success complement simulator success; existing task traces do not yet supply every comparable aggregate. A shared efficiency panel should measure encoder-inclusive and predictor-only latency, complete CEM replanning cost where applicable, peak memory, active parameters and training GPU-hours under matched hardware and precision. These are pending measurements, not zero-valued results. RGB/video scores require a separately evaluated decoder; they cannot be inferred from feature errors. New numerical cells and plots are populated from completed, source-bound comparisons. Ongoing single-seed diagnostics do not bypass the full-campaign reporting gate. All unfavorable results and prespecified conditions remain visible.

## B OBSERVATION-ANCHORED SPATIAL DEVELOPMENT STUDY

This registered architecture study addresses recursive prediction drift while preserving the locations of visual features. It is a new spatial variant; it does not preserve the original separate observation/dynamics factorization. The frozen DINOv2-small encoder produces a  $4 \times 4$  grid of 384-dimensional features. Training-only channel statistics are shared across all sixteen positions, so mixtures remain in common feature coordinates. Three observed grids pass through a shared spatial encoder and the pinned LeWM temporal predictor. A causal action-prefix GRU supplies commands only through the requested horizon, and a support-only transition context conditions the trunk through FiLM. Context remains fixed throughout the forecast.

For horizon  $h$ , let  $Z_0$  be the last actual observed grid,  $H_h$  the horizon state, and  $E_0$  the spatial encoding of  $Z_0$  before context FiLM. The transport decoder computes

$$T_h = \text{softmax}_{\text{row}} \left( Q(H_h) K(E_0)^\top / \sqrt{96} + 4I \right), \quad (3)$$

$$\hat{Z}_h = (1 - g_h) \odot Z_0 + g_h \odot (T_h Z_0) + \tanh(R(H_h)), \quad g_h = \sigma(W_g H_h + b_g). \quad (4)$$

Every forecast retains the actual anchor instead of recursively replacing it with predictions. The row-stochastic mixture and unit-bounded innovation imply the componentwise bound  $|\hat{Z}_{h,p,c}| \leq \max_q |Z_{0,q,c}| + 1$  in shared normalized coordinates. This is an output invariant, not an accuracy or stability guarantee. Transport mixes semantic features; it is not a validated physical image warp or RGB generator.

The five registered arms share the spatial inputs and trunk: Autoregressive recursively updates the feature history; Anchored additive keeps the actual observation anchor; Spatial transport (ours) adds mixing and bounded innovation; Context-off and Action-free disable the corresponding information paths. Transport begins near persistence (identity score bias 4 and gate bias  $-3$ ), whereas the zero-initialized additive heads begin at exact persistence. Active parameter counts differ slightly. These arms test the package-level decoder effect; they do not separately identify mixing versus bounded innovation.

All fifteen models complete 30 epochs at seeds 0/1/2 with identical AdamW settings, batch size 128 and ten query steps. Window-weighted all-ten validation MSE selects checkpoints; final metrics average windows within episodes, then weight episodes and seeds equally. The evaluation contains 141 episodes from 59 recording sessions and 1631 windows. Neither original nor fresh held-out test observations enter this development study.

Table 5 brings all eight predictors, including the completed follow-up component arms, into one native/pooled scorecard. Each coordinate system has its own normalization; comparisons are made within columns. Tables 6 and 7 preserve all sixteen registered comparisons, including regressions and intervals containing zero. These intervals are exploratory and unadjusted for multiple comparisons; stronger claims require an independently frozen evaluation. Table 8 records every selected checkpoint and the nonidentical active parameter counts.

Table 5: Complete current DROID spatial-development scorecard. Lower feature MSE is better; black bold marks the lowest point mean in each column, not significance. Native and pooled coordinates use different fixed training normalizations and cannot be compared numerically across groups. All seven learned variants completed 30 epochs at three seeds; persistence is untrained. Means weight windows within episode, then 141 episodes and seeds equally (59 sessions; 1,631 windows). Each query step spans five native transitions. Feature MAE/cosine, RGB quality and physical-control success were not measured in this spatial study. Paired uncertainty remains in the spatial comparison tables; the source pack retains every seed curve.

| Method                           | Native $4 \times 4$ MSE $\downarrow$ |                 | Pooled $2 \times 2$ MSE $\downarrow$ |                 |
|----------------------------------|--------------------------------------|-----------------|--------------------------------------|-----------------|
|                                  | h5                                   | h10             | h5                                   | h10             |
| Autoregressive                   | 0.149573                             | 0.203950        | 0.148577                             | 0.202916        |
| Persistence                      | 0.164028                             | 0.226129        | 0.163156                             | 0.225622        |
| ShiftWM (ours)                   | 0.145074                             | <b>0.193141</b> | 0.147921                             | 0.197766        |
| No mixing (ours, ablation)       | 0.149347                             | 0.200010        | 0.148430                             | 0.198261        |
| No bound (ours, ablation)        | <b>0.145037</b>                      | 0.193274        | 0.147812                             | 0.197706        |
| Additive anchor (ours, ablation) | 0.149606                             | 0.200244        | 0.148705                             | 0.198704        |
| No context (ours, ablation)      | 0.145143                             | 0.193330        | <b>0.147635</b>                      | <b>0.197553</b> |
| No actions (ours, ablation)      | 0.149480                             | 0.202249        | 0.152414                             | 0.206188        |

## B.1 WHAT THE SPATIAL CONTROLS ESTABLISH

At the ten-step endpoint, Spatial transport (ours) reduces native feature MSE by 5.30% relative to the matched autoregressive predictor (0.193141 versus 0.203950) and by 3.55% relative to Anchored additive (0.200244). Both paired intervals exclude zero. These are relative reductions in feature error, not percentage-point improvements in task success. On the original pooled coordinates, the reduction is 2.54% against autoregression, while the 0.47% difference against anchoring has an interval containing zero. The stronger native-grid result therefore does not establish an equally strong advantage at every spatial resolution.

Removing the command-conditioning path increases native ten-step error by the equivalent of a 4.50% relative reduction for the full model. In contrast, the separate support-context path adds only a 0.10% native endpoint point gain; all four context comparisons have intervals containing zero, including two small unfavorable pooled-coordinate differences. These controls provide evidence for the action-conditioned decoder package, but do not establish an additional benefit from its learned support context. The action-free and context-off controls also remove active parameters: context-

| Transport comparator | $h$ | Gain                        | Difference [paired 95% interval] |
|----------------------|-----|-----------------------------|----------------------------------|
| Autoregressive       | 5   | <b>+3.008%</b>              | -0.004499 [-0.006132, -0.002491] |
| Autoregressive       | 10  | <b>+5.300%</b>              | -0.010809 [-0.012461, -0.009028] |
| Anchored additive    | 5   | <b>+3.029%</b>              | -0.004532 [-0.005930, -0.002795] |
| Anchored additive    | 10  | <b>+3.547%</b>              | -0.007102 [-0.008831, -0.005019] |
| Context-off          | 5   | <b>+0.048%</b> <sup>†</sup> | -0.000069 [-0.000633, +0.000478] |
| Context-off          | 10  | <b>+0.097%</b> <sup>†</sup> | -0.000188 [-0.000943, +0.000485] |
| Action-free          | 5   | <b>+2.948%</b>              | -0.004406 [-0.006551, -0.002149] |
| Action-free          | 10  | <b>+4.503%</b>              | -0.009107 [-0.011896, -0.006061] |

Table 6: **Transport comparisons: Native  $4 \times 4$  features.** Differences are Spatial transport (ours) minus comparator; negative favors ours. All four comparators and both endpoints are included. Intervals resample recording sessions and three training seeds jointly (10,000 draws; seed 173). <sup>†</sup> marks intervals containing zero. Green bold denotes positive point gain, not significance. Exploratory development evidence, without multiple-comparison adjustment.

| Transport comparator | $h$ | Gain                        | Difference [paired 95% interval] |
|----------------------|-----|-----------------------------|----------------------------------|
| Autoregressive       | 5   | <b>+0.441%</b> <sup>†</sup> | -0.000656 [-0.002499, +0.001918] |
| Autoregressive       | 10  | <b>+2.538%</b>              | -0.005150 [-0.007196, -0.002727] |
| Anchored additive    | 5   | <b>+0.527%</b> <sup>†</sup> | -0.000784 [-0.002281, +0.001370] |
| Anchored additive    | 10  | <b>+0.472%</b> <sup>†</sup> | -0.000938 [-0.002897, +0.001784] |
| Context-off          | 5   | -0.194% <sup>†</sup>        | +0.000286 [-0.000386, +0.001016] |
| Context-off          | 10  | -0.108% <sup>†</sup>        | +0.000213 [-0.000644, +0.001075] |
| Action-free          | 5   | <b>+2.948%</b>              | -0.004493 [-0.007086, -0.001778] |
| Action-free          | 10  | <b>+4.085%</b>              | -0.008422 [-0.011892, -0.004691] |

Table 7: **Transport comparisons: Original  $2 \times 2$  coordinates.** Differences are Spatial transport (ours) minus comparator; negative favors ours. All four comparators and both endpoints are included. Intervals resample recording sessions and three training seeds jointly (10,000 draws; seed 173). <sup>†</sup> marks intervals containing zero. Green bold denotes positive point gain, not significance. Exploratory development evidence, without multiple-comparison adjustment.

| Method                          | Selected epochs | Total params | Active params |
|---------------------------------|-----------------|--------------|---------------|
| Autoregressive                  | 16/25/22        | 1,026,305    | 1,007,776     |
| Anchored additive               | 15/25/16        | 1,026,305    | 1,007,776     |
| <b>Spatial transport (ours)</b> | 14/19/11        | 1,026,305    | 1,026,305     |
| Context-off                     | 10/17/14        | 1,026,305    | 878,497       |
| Action-free                     | 6/5/4           | 1,026,305    | 988,001       |

Table 8: **All fifteen completed spatial models.** Every model completed 30 epochs; selected epochs follow seeds 0/1/2 and minimize window-weighted all-ten validation error. Active counts exclude frozen unused heads. The shared trunk does not imply identical active parameter counts; Context-off and Action-free remove inputs and parameters. All selected models passed relocated CPU forecast parity.

| ID     | Mechanism                             | Status                        |
|--------|---------------------------------------|-------------------------------|
| Ours-1 | Original context-based ShiftWM        | Completed; different protocol |
| Ours-2 | Observed anchor + additive innovation | Completed                     |
| Ours-3 | Observed anchor + bounded innovation  | Completed; follow-up          |
| Ours-4 | Gated mixing + unbounded innovation   | Completed; follow-up          |
| Ours-5 | Gated mixing + bounded innovation     | Completed                     |

Table 9: Stable mechanism identifiers, not a performance ranking. Ours-1 uses the original context-based protocol and is excluded from the spatial comparison axis. Context-off and action-free are Ours-5 ablations; autoregression and persistence retain baseline labels. Ours-3/4 completed their registered follow-up after the original controls were revealed; all six new models trained 30 epochs.

off has 878,497 versus 1,026,305 for the full model (14.40% fewer). They are information-path ablations, not exact capacity matches.

Across sixteen registered contrasts, fourteen point estimates favor transport, nine intervals exclude zero in that direction, and seven intervals contain zero. Two of the latter have unfavorable point estimates. Because all comparisons use development validation data and unadjusted intervals, this study supports further architectural investigation rather than a new confirmatory superiority claim.

**Separating mixing and bounding.** A registered follow-up completes the two-by-two decoder comparison with six additional 30-epoch models: bounded additive (Ours-3) and unbounded transport (Ours-4), each with three seeds. Together with the six previously revealed anchored-additive and bounded-transport controls, this isolates conditional component comparisons under the same training recipe. At the native ten-step endpoint, adding the mixing package reduces error by 3.48% without bounding and 3.43% with bounding; both paired intervals exclude zero. In contrast, the incremental native-coordinate effect of bounding is inconclusive at both reported horizons, and all four mixing-by-bounding interaction intervals contain zero. Two small pooled-coordinate bounding effects without mixing have favorable intervals; the complete table retains them alongside every null and unfavorable point estimate.

These results support the action-conditioned mixing package as the main measured improvement over the additive anchor in this development study. They do not establish an accuracy advantage from the bounded correction itself. The package includes its learned gate and identity bias and adds approximately 1.84% active parameters, so its comparison does not isolate mixing from those accompanying changes. Ours-4 and Ours-5 trade small point differences across metrics; version numbers are mechanism identifiers, not a monotonic performance ranking. The six new inference packages pass exact relocated CPU reload checks.

## B.2 COMPONENT ABLATIONS AND HISTORICAL IDENTIFIERS

The proposed spatial model is the transport arm, historically labeled Ours-5. Ours-2–4 are component controls, not independently proposed algorithms; Ours-1 denotes the earlier, distinct two-context model. Stable identifiers below preserve checkpoint and result-file correspondence. Their numbering never implies a performance ranking. All spatial arms have three-seed results.

## ShiftWM (ours) versus matched spatial controls



Figure 5: **Paired native-error reductions for ShiftWM (ours) and its controls.** Both panels report comparator minus ShiftWM endpoint MSE in the native  $4 \times 4$  standardized feature space; right of zero favors ShiftWM. Percentages are relative reductions against each named comparator. Bold green identifies positive point gains, not significance; open marks have intervals crossing zero. Bars are paired 95% recording-session  $\times$  training-seed bootstrap intervals (10,000 draws, seed 173), without multiplicity adjustment. The slight h5 disadvantage against the unbounded model and all inconclusive effects remain visible. The eight absolute method scores are in Table 1; Tables 10 and 11 retain all 36 native/coarse-coordinate effects and signed interactions. All 21 models completed 30 epochs, including six follow-up models added after the original controls were revealed. The matched development population contains 1,631 windows from 141 episodes, 59 sessions and three seeds, averaging windows within episodes and then episodes and seeds equally. These are ablations of one proposed method, not separate algorithms or independent-test claims.

| Metric                | Comparator       | $h$ | Relative gain  | MSE reduction $\times 10^3$ [95% CI] |
|-----------------------|------------------|-----|----------------|--------------------------------------|
| Native                | Autoregressive   | 5   | <b>+3.008%</b> | +4.499 [+2.491, +6.132]              |
| Native                | Autoregressive   | 10  | <b>+5.300%</b> | +10.809 [+9.028, +12.461]            |
| Native                | Ours-2           | 5   | <b>+3.029%</b> | +4.532 [+2.795, +5.930]              |
| Native                | Ours-2           | 10  | <b>+3.547%</b> | +7.102 [+5.019, +8.831]              |
| Native                | Ours-5 – context | 5   | <b>+0.048%</b> | +0.069 [-0.478, +0.633]              |
| Native                | Ours-5 – context | 10  | <b>+0.097%</b> | +0.188 [-0.485, +0.943]              |
| Native                | Ours-5 – actions | 5   | <b>+2.948%</b> | +4.406 [+2.149, +6.551]              |
| Native                | Ours-5 – actions | 10  | <b>+4.503%</b> | +9.107 [+6.061, +11.896]             |
| Original $2 \times 2$ | Autoregressive   | 5   | <b>+0.441%</b> | +0.656 [-1.918, +2.499]              |
| Original $2 \times 2$ | Autoregressive   | 10  | <b>+2.538%</b> | +5.150 [+2.727, +7.196]              |
| Original $2 \times 2$ | Ours-2           | 5   | <b>+0.527%</b> | +0.784 [-1.370, +2.281]              |
| Original $2 \times 2$ | Ours-2           | 10  | <b>+0.472%</b> | +0.938 [-1.784, +2.897]              |
| Original $2 \times 2$ | Ours-5 – context | 5   | -0.194%        | -0.286 [-1.016, +0.386]              |
| Original $2 \times 2$ | Ours-5 – context | 10  | -0.108%        | -0.213 [-1.075, +0.644]              |
| Original $2 \times 2$ | Ours-5 – actions | 5   | <b>+2.948%</b> | +4.493 [+1.778, +7.086]              |
| Original $2 \times 2$ | Ours-5 – actions | 10  | <b>+4.085%</b> | +8.422 [+4.691, +11.892]             |

Table 10: All 16 completed spatial contrasts: Ours-5 versus each comparator. Positive reductions favor Ours-5; bold green percentages indicate positive point estimates, not statistical significance. Paired 95% recording-session and seed bootstrap intervals use 10,000 draws (seed 173), without multiplicity correction. Both unfavorable coarse-coordinate context-ablation point estimates remain included. Native and original  $2 \times 2$  coordinate errors are different metrics and cannot be pooled. Original validation only.

| Comparison       | Metric       | $h$ | Relative gain  | Effect $\times 10^3$ [95% CI] |
|------------------|--------------|-----|----------------|-------------------------------|
| Ours-3 vs Ours-2 | Native       | 5   | <b>+0.173%</b> | +0.259 [-0.017, +0.535]       |
| Ours-3 vs Ours-2 | Native       | 10  | <b>+0.117%</b> | +0.234 [-0.160, +0.649]       |
| Ours-3 vs Ours-2 | $2 \times 2$ | 5   | <b>+0.185%</b> | +0.275 [+0.009, +0.533]       |
| Ours-3 vs Ours-2 | $2 \times 2$ | 10  | <b>+0.223%</b> | +0.443 [+0.009, +0.907]       |
| Ours-5 vs Ours-4 | Native       | 5   | -0.025%        | -0.037 [-0.387, +0.309]       |
| Ours-5 vs Ours-4 | Native       | 10  | <b>+0.069%</b> | +0.133 [-0.263, +0.606]       |
| Ours-5 vs Ours-4 | $2 \times 2$ | 5   | -0.074%        | -0.109 [-0.559, +0.287]       |
| Ours-5 vs Ours-4 | $2 \times 2$ | 10  | -0.030%        | -0.060 [-0.652, +0.449]       |
| Ours-4 vs Ours-2 | Native       | 5   | <b>+3.054%</b> | +4.569 [+2.917, +5.870]       |
| Ours-4 vs Ours-2 | Native       | 10  | <b>+3.481%</b> | +6.970 [+5.013, +8.666]       |
| Ours-4 vs Ours-2 | $2 \times 2$ | 5   | <b>+0.601%</b> | +0.893 [-1.162, +2.359]       |
| Ours-4 vs Ours-2 | $2 \times 2$ | 10  | <b>+0.502%</b> | +0.998 [-1.653, +2.923]       |
| Ours-5 vs Ours-3 | Native       | 5   | <b>+2.861%</b> | +4.273 [+2.542, +5.650]       |
| Ours-5 vs Ours-3 | Native       | 10  | <b>+3.434%</b> | +6.868 [+4.723, +8.690]       |
| Ours-5 vs Ours-3 | $2 \times 2$ | 5   | <b>+0.343%</b> | +0.509 [-1.618, +1.935]       |
| Ours-5 vs Ours-3 | $2 \times 2$ | 10  | <b>+0.250%</b> | +0.495 [-2.224, +2.513]       |
| Interaction $D$  | Native       | 5   | —              | +0.296 [-0.115, +0.778]       |
| Interaction $D$  | Native       | 10  | —              | +0.101 [-0.471, +0.710]       |
| Interaction $D$  | $2 \times 2$ | 5   | —              | +0.384 [-0.061, +0.938]       |
| Interaction $D$  | $2 \times 2$ | 10  | —              | +0.503 [-0.175, +1.344]       |

Table 11: All 20 predeclared component effects from the original-validation follow-up: six new models plus six reused controls, each trained for 30 epochs. The first four blocks report error reductions (second-model MSE minus first-model MSE): positive favors the first model. Bounding is isolated within the Ours-3/Ours-2 and Ours-5/Ours-4 pairs; mixing is compared within Ours-4/Ours-2 and Ours-5/Ours-3. The last block retains the separately signed interaction  $D = (\text{MSE}_5 - \text{MSE}_4) - (\text{MSE}_3 - \text{MSE}_2)$ : negative means bounding helps more with mixing; no percentage is defined for  $D$ . All four interaction intervals include zero. Bold green denotes positive pairwise point estimates, not significance. Paired session/seed 95% bootstrap intervals use 10,000 draws, seed 173, without multiplicity correction. This follow-up was designed after existing controls were revealed; mixing includes the gate, identity bias and approximately 1.84% extra active parameters.

### B.3 MEASURED DECODER DIAGNOSTICS

Figure 2 separates model inputs from withheld evaluation. In the illustrated DROID window, observed frames have source indices 0, 5 and 10; each action block spans five native source-frame intervals, so the  $h = 10$  target is frame 60. These indices denote frames, not seconds. Figure 4 compares the measured forecast errors. Panel (a) of Figure 6 compares the decoder’s internal weights across four recorded-video settings. Its DROID example is the first eligible window of the median episode by episode-level gain, retaining the original selection from Figure 4; its three seeds are 0, 1 and 2. Each IWS example uses training trajectory 000011, initial frame 0, and command rows 0–59, fixed before inspecting these decoder traces. The checkpoint snapshot contains PushT seeds 0/1, Box seeds 0/1/2 and Rope seeds 0/2; every included checkpoint completed the full 30-epoch training budget and passed its individual development-evaluation checks. The other two mixing seeds were unfinished at this snapshot. This descriptive replay is separate from the complete-campaign comparison gate and is not evidence of unseen-data generalization. The initial IWS observation is repeated into the three temporal input slots, and support-context conditioning is disabled. The maps preserve each patch’s own-source weight, with scalar means across all 16 patches; the archived three-case DROID replay also retains the full source-weight matrices, observations and predictions. Neither diagnostic establishes physical correspondence or a causal explanation of forecast error. Panel (b) gives complementary forecast results for the historical context model (Appendix I), retaining unfavorable outcomes. These separate simulator experiments do not evaluate the current spatial checkpoints. Full gate and retention trajectories remain in the source archive.



Figure 6: **Spatial source retention and historical simulator forecasts.** (a) For ShiftWM’s spatial decoder, each cell is one patch’s own-source weight: the diagonal of raw mixing  $T$  or effective mixing  $M = I - D + DT$ . All eight maps share a linear  $[0, 1]$  scale. Means below the maps average 16 patches and the displayed seeds, after forming  $M$  per seed. Endpoints are DROID step 10 and IWS stored offset 59; case selection and training/development scope are specified above. These source weights precede the bounded correction. Their increase is a construction property, not an accuracy gain. (b) The separate historical context model is compared with Framewise calibration using fixed-reference MSE@5. Pair/Extra denote held-out composition and the three-condition extrapolation aggregate; Trf./GRU denote both predictor architectures on drone/tissue development data. Positive values mean lower error. Dark markers and labels use the ratio of three-seed mean MSEs; pale dots are seedwise ratios. All stems share a  $[-10, 30]$  percent scale with a tick at zero; no confidence interval is implied. Simulator frames identify tasks, not scored examples.

## C EXPERIMENT ALIGNMENT AND SINGLE-OBSERVATION TRANSFER

The current proposed method is ShiftWM’s bounded observation-anchored spatial decoder. The DROID development study evaluates its three-observation instantiation. The historical context

model and the single-observation transfer study have different interfaces and selection protocols; their scores must not be pooled into one cross-domain superiority claim.

Table 12: Study interfaces and evidence status. DROID results use development validation; IWS development results use the completed, verified single-observation protocol; official evaluation remains separate. Historical context-model protocols remain separate.

| Task      | Inputs and outputs                                                                | Score and comparison                                                       | Status                                    |
|-----------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------|
| DROID     | Three images; two past and ten query action blocks; future $4 \times 4$ features. | Native $h = 10$ standardized MSE; matched AR and additive-anchor controls. | 21 models complete; development only.     |
| IWS PushT | One image; 60 native 4D command rows; 59 future feature grids.                    | $H = 60$ standardized MSE; additive anchor is the primary comparator.      | 600 cached; matched development complete. |
| IWS Box   | One image; 60 native 14D command rows; 59 future feature grids.                   | $H = 60$ standardized MSE; additive anchor is the primary comparator.      | 602 cached; matched development complete. |
| IWS Rope  | One image; 60 native 8D command rows; 59 future feature grids.                    | $H = 60$ standardized MSE; additive anchor is the primary comparator.      | 602 cached; matched development complete. |

Table 13: Complete IWS internal-development endpoints at  $H = 60$ . MSE averages windows within trajectory, then trajectories and matched seeds equally. Gains and paired MSE-difference intervals compare ShiftWM with the additive anchor. Positive gains are bold green; intervals are unadjusted and may include zero. All three tasks, three learned arms, three seeds and persistence are retained. Official evaluation is not included.

| Task  | Method          | MSE      | Gain vs anchor | Paired 95% CI        |
|-------|-----------------|----------|----------------|----------------------|
| PushT | Persistence     | 0.571659 | —              | —                    |
| PushT | Autoregressive  | 0.259999 | —              | —                    |
| PushT | Additive anchor | 0.267429 | —              | —                    |
| PushT | ShiftWM (ours)  | 0.261116 | <b>2.360%</b>  | [-0.00700, -0.00563] |
| Box   | Persistence     | 0.630393 | —              | —                    |
| Box   | Autoregressive  | 0.279410 | —              | —                    |
| Box   | Additive anchor | 0.296560 | —              | —                    |
| Box   | ShiftWM (ours)  | 0.290927 | <b>1.900%</b>  | [-0.00720, -0.00413] |
| Rope  | Persistence     | 0.617137 | —              | —                    |
| Rope  | Autoregressive  | 0.211976 | —              | —                    |
| Rope  | Additive anchor | 0.222888 | —              | —                    |
| Rope  | ShiftWM (ours)  | 0.222913 | -0.011%        | [-0.00130, +0.00152] |

Equal-task macro relative reduction against the additive anchor: 1.416%, paired 95% interval [+1.147, +1.675]. Table 13 reports intervals for the MSE difference, ShiftWM minus the additive anchor; negative differences favor ShiftWM. Table 2 and Figure 7 instead report intervals for relative error reduction in percent, where positive values favor ShiftWM. Both use the same paired resampling draws.

**Single-observation transfer contract.** The IWS interface takes one observed image and  $H$  recorded native command rows, then predicts the  $H - 1$  subsequent stored-frame features. For  $H = 60$ , the observed index is  $s$ , the target is  $s + 59$ , and all command rows  $s, \dots, s + 59$  condition the endpoint. These are stored indices, not verified seconds or sixty physical transitions. The three temporal slots initially repeat the one observed feature grid; they do not represent three observed frames. Support-transition context and FiLM are disabled because one image supplies no observed transition.

The study grid specifies separate per-task checkpoints for autoregression, additive anchoring and bounded spatial mixing, with three seeds each; persistence requires no training. PushT, Box and

Rope retain their native command widths of 4, 14 and 8. Internal splits contain 480/120, 481/121 and 481/121 training/development trajectories, respectively. Task-specific training-only statistics and the full-RGB DINOv2 feature representation are shared between arms within each task. This is a transfer of the decoder mechanism, not a shared cross-embodiment checkpoint or an RLA-WM reproduction.

**Selection, primary comparison and reserved evaluation.** The full-horizon recipe trains all 59 offsets and selects checkpoints by internal-development equal-trajectory endpoint MSE at  $H = 60$ . The declared primary comparison is bounded mixing versus the additive anchor, with task-specific standardized feature-MSE reductions and an equal-task macro of those reductions. Autoregression, persistence, shorter endpoints and other feature metrics are secondary. The upstream reserved set contains 200 handles from ten trajectories per task; a handle is not an independent trajectory. Original and derived prefix endpoints remain separately labeled. The exact architecture, recipe, metric aggregation, uncertainty procedure and checkpoint identities must be locked before these reserved payloads are accessed. The status table distinguishes prepared data, execution status and completed measurements; the interface alone is not performance evidence.

**Reusable IWS predictors.** All 27 selected checkpoints have inference-only exports containing their exact weights, task-specific statistics and frozen model code. A relocated, offline CPU process reproduces every forecast offset exactly on a fixed training input for each task. This checks portability, not additional forecasting accuracy. Inputs are one raw 6,144-dimensional DINOv2 feature vector and native commands; outputs are future feature vectors. The local packages exclude optimizer state, encoder weights, RGB decoders and future-target inputs. Their complete manifests and export instructions accompany the source release; checkpoint uploads remain separate. CPU parity does not certify the CUDA backend.

**Result ingestion boundary.** Only the complete 27-run finalizer can populate numerical cells. It verifies the full training budget, checkpoint and source identities, every eligible window and independently reconstructed metrics. Development and reserved evaluation use distinct finalizers. Negative and inconclusive results remain visible; incomplete cells remain pending. IWS development numbers are generated only from all 27 completed, source-bound evaluation receipts; official evaluation is not ingested.



Figure 7: **Complete IWS development forecasts and the innovation-bound ablation.** All five predictors retain all 59 future offsets from one observed image; learned methods receive supplied causal command prefixes, while persistence ignores commands. The horizontal axis is  $H = 2, \dots, 60$ , targeting stored offset  $H - 1$ , not physical time. Training-standardized feature MSE weights windows within trajectory, trajectories and three seeds equally; lower is better. Each task has its own zero-based vertical scale. The no-tanh predictor removes only the correction bound and belongs to a separately registered exploratory follow-up after v1 development. All nine new and 27 original runs must pass finalization before this figure appears. No uncertainty bands are inferred; paired endpoint gains and intervals are in Table 15. Reserved upstream-validation data remain excluded.

Table 14: Complete IWS internal-development feature metrics at  $H = 60$  (stored offset 59). Lower is better in every column; black bold marks each task/metric's lowest point mean, not significance. Learned methods use one image and supplied causal commands; persistence ignores commands. Means weight windows within trajectory, trajectories and three seeds equally. MSE/MAE use task-specific training scales; raw L1 uses DINOv2 coordinates; cosine distance uses flattened raw features. RGB quality and physical-control success were not measured. Reserved upstream-validation data are excluded. The main MSE comparison table retains paired gains and intervals. The no-tanh row is a separately registered exploratory follow-up after v1 development, included only after all nine new and 27 original runs passed complete-study validation.

| Method                                         | Std. MSE ↓      | Std. MAE ↓      | Raw L1 ↓        | Cosine ↓        |
|------------------------------------------------|-----------------|-----------------|-----------------|-----------------|
| <b>PushT</b> — 120 trajectories, 3,360 windows |                 |                 |                 |                 |
| Persistence                                    | 0.571659        | 0.566375        | 0.602920        | 0.103964        |
| Autoregressive                                 | 0.259999        | 0.392734        | 0.415426        | 0.045451        |
| Additive anchor                                | 0.267429        | 0.396837        | 0.418767        | 0.046569        |
| ShiftWM (ours)                                 | 0.261116        | 0.390369        | 0.411517        | 0.045501        |
| No tanh (ours, ablation)                       | <b>0.241618</b> | <b>0.376721</b> | <b>0.396830</b> | <b>0.041825</b> |
| <b>Box</b> — 121 trajectories, 3,388 windows   |                 |                 |                 |                 |
| Persistence                                    | 0.630393        | 0.596194        | 0.622461        | 0.105339        |
| Autoregressive                                 | 0.279410        | 0.405636        | 0.424386        | 0.047147        |
| Additive anchor                                | 0.296560        | 0.416768        | 0.435191        | 0.049976        |
| ShiftWM (ours)                                 | 0.290927        | 0.410230        | 0.427627        | 0.048972        |
| No tanh (ours, ablation)                       | <b>0.263285</b> | <b>0.392340</b> | <b>0.408863</b> | <b>0.044062</b> |
| <b>Rope</b> — 121 trajectories, 3,388 windows  |                 |                 |                 |                 |
| Persistence                                    | 0.617137        | 0.585841        | 0.603724        | 0.102943        |
| Autoregressive                                 | 0.211976        | 0.354678        | 0.362410        | 0.033471        |
| Additive anchor                                | 0.222888        | 0.363139        | 0.370141        | 0.035015        |
| ShiftWM (ours)                                 | 0.222913        | 0.359899        | 0.365638        | 0.034708        |
| No tanh (ours, ablation)                       | <b>0.197046</b> | <b>0.340392</b> | <b>0.345662</b> | <b>0.030523</b> |

Table 15: Post-development IWS innovation-bound ablation. Nine new 30-epoch runs are compared with all 27 frozen common-CPU v1 runs. Only tanh is removed from the correction. Entries are  $H = 60$  standardized MSE (stored offset 59), weighting windows within trajectory, trajectories and three seeds equally; lower is better. Gains and paired 95% intervals are percentages versus bounded ShiftWM; positive values favor removing the bound. Intervals use 10,000 paired seed–trajectory draws. These unadjusted exploratory comparisons were designed after v1 development. All tasks, controls and signs remain visible; reserved validation is excluded.

| Task  | Persistence | AR       | Additive<br>anchor | ShiftWM<br>(ours) | No tanh<br>(ablation) | Gain [95% CI]<br>vs ShiftWM    |
|-------|-------------|----------|--------------------|-------------------|-----------------------|--------------------------------|
| PushT | 0.571659    | 0.259999 | 0.267429           | 0.261116          | 0.241618              | <b>+7.47</b> [+7.08, +7.85]    |
| Box   | 0.630393    | 0.279410 | 0.296560           | 0.290927          | 0.263285              | <b>+9.50</b> [+8.92, +10.08]   |
| Rope  | 0.617137    | 0.211976 | 0.222888           | 0.222913          | 0.197046              | <b>+11.60</b> [+11.11, +12.10] |

Equal-task mean relative gain: +9.52%; paired 95% interval [+9.24, +9.80]%.

## D REGISTERED STUDY ON RECORDED ROBOT VIDEO

**Status: complete prespecified real-data comparison.** This additional study evaluates forecasting on real robot recordings from DROID (Khazatsky et al., 2024). It complements the simulator experiments; it does not replace them or establish real-robot control success. Figure 8 follows one recorded episode from observed history and logged commands to future-feature forecasting and withheld-target scoring. Its camera panel separates the primary view from the transfer test.

**Data and splits.** Before optimization, we fixed 24 of the full release’s 2,048 shards by a salted hash of their object names. This selection comprises 1,126 audited episodes and 22,017,792,821 downloaded bytes including metadata and license. Object generations, publisher checksums and local hashes identify the inputs. We assign recording sessions, defined by site plus recording date, to training, validation and test with fixed 70/15/15 hash thresholds. The resulting splits contain 851, 143 and 132 episodes from 316, 59 and 58 disjoint sessions, respectively, spanning 14 sites overall. All 327,498 native observations and 64,826 complete action blocks passed extraction and alignment checks. Unparseable session identifiers fail the audit; there is no frame-level fallback. Session separation does not guarantee distinct scenes or objects. All recorded outcomes are retained without success-based selection.

**Observation and action contract.** Frames at native indices 0, 5, 10, ... are paired with the intervening five seven-dimensional commands, concatenated into a 35-dimensional block. The full extraction verified every native command array against the recorded six Cartesian-position/pose components plus gripper position. Incomplete tails are counted, and 11 episodes too short for an eight-frame training window remain in the data. The inspected release has no per-step timestamp, so horizons are expressed in native transitions. Ground-truth state, rewards, success flags, language and future images are never context inputs. Training and checkpoint selection use external camera 1 only; camera 2 on held-out episodes provides a separately reported natural camera-transfer test. No synthetic appearance corruption is applied in this study.

**Representation and matched controls.** We freeze DINOv2-small (Oquab et al., 2023), resize complete images to  $224 \times 224$ , and pool the final patch grid into  $2 \times 2$  spatial cells, producing 1,536-dimensional targets. Feature and action normalization is fitted on training camera 1 only. The pinned LeWM causal predictor (Maes et al., 2026b) has width 192 and four transformer blocks; a residual output head initializes every mode to persistence in the fixed feature space. The four modes are Framewise, Constant dynamics, ShiftWM real-video variant (ours), and an Action-free control. Each uses three training seeds and the same common initialization, data and optimization: 12 registered runs, each with 30 epochs. The model observes three support frames and recursively predicts five further frames using recorded commands, without query-image teacher forcing. We select checkpoints using validation forecast loss.

**Evaluation and interpretation.** We register horizons of 1, 3 and 5 action blocks, plus a separately labeled 10-block extrapolation test. Comparisons include persistence and constant feature-velocity extrapolation, with raw and standardized feature MSE and cosine error. We aggregate within episodes and quantify uncertainty by source session and training seed. Camera-transfer and ordinary held-out results remain separate. Reversing recorded query actions probes model sensitivity, not the physical outcome of an unexecuted intervention. This observational study provides neither paired canonical targets nor identified physical factors; its metrics cannot establish closed-loop real-robot success.



**Figure 8: From recorded robot interaction to a forecasting test.** (A) Camera 1 supplies training, validation and primary testing; camera 2 is used in a separate held-out viewpoint-transfer test. The wrist camera is retained for inspection. Camera placement is schematic, with one camera stream per model run. (B) Three recorded observations at native frames 0, 5 and 10, two past command blocks and five supplied future blocks condition an offline forecast. Each block contains five seven-dimensional recorded commands. Five successive future feature vectors are predicted; the drawing shows the fifth, at frame 35. (C) The corresponding recorded future image enters only the evaluator. Both paths use the same frozen DINOv2-small encoder,  $2 \times 2$  pooling and training-camera-1 feature statistics. Predictions are compared with the fixed 1,536-dimensional targets using feature MSE and cosine error. The section reports all registered horizons and both camera populations separately. Photos show the first eligible test episode by lexicographic ID, selected independently of outcomes. All views retain their full frame and publisher-provided blurring. The camera illustration and feature tiles are schematic, while photographs are unchanged DROID observations (CC BY 4.0). This is the historical context-model study; the current spatial architecture is in Figure 2.

**Completed results.** All 12 runs completed 30 epochs. On the primary camera, five-block standardized feature MSE is 0.138198 for ShiftWM, 0.138479 for Framewise, and 0.142392 for persistence. The 0.20% reduction versus Framewise is inconclusive: the paired 95% MSE-difference interval is  $[-0.001038, 0.000413]$ . The 2.94% reduction versus persistence has an interval below zero,  $[-0.006714, -0.001908]$ ; camera transfer retains a 2.02% reduction versus persistence. At ten blocks, however, error is 1.91% and 1.24% higher than Framewise on the two cameras, respectively, with both intervals including zero. These intervals are unadjusted for multiple comparisons. Figure 9A relates recorded observations to the measured episode-level errors.

**Training and action-use limitations.** Every validation-selected checkpoint occurs at epoch 1 or 2; later validation deterioration despite decreasing training loss indicates overfitting. Reversing the order of future action blocks changes the final-horizon error of ShiftWM by less than 0.10% in every population. This limited sensitivity is an observational diagnostic, not proof that actions are unused or that physical interventions have no effect. The completed study supports a modest short-horizon benefit over persistence, while a substantial advantage over matched learned baselines and reliable long-horizon action-dependent prediction remain unresolved.

| Method                | MSE@1 ↓                          | MSE@3 ↓                       | MSE@5 ↓              | Gain@5 (%) ↑ |
|-----------------------|----------------------------------|-------------------------------|----------------------|--------------|
| Framewise             | $0.04988 \pm 2.2 \times 10^{-5}$ | $0.1031 \pm 4 \times 10^{-5}$ | $0.1385 \pm 0.00015$ | reference    |
| Constant dynamics     | $0.04987 \pm 2.2 \times 10^{-5}$ | $0.1031 \pm 5 \times 10^{-5}$ | $0.1384 \pm 0.00016$ | +0.04        |
| <b>ShiftWM (ours)</b> | $0.04979 \pm 4.2 \times 10^{-5}$ | $0.1029 \pm 0.00011$          | $0.1382 \pm 0.00011$ | <b>+0.20</b> |
| Action-free           | $0.04993 \pm 2.8 \times 10^{-5}$ | $0.1033 \pm 0.00011$          | $0.1389 \pm 0.00013$ | -0.28        |
| Persistence           | 0.05029                          | 0.1053                        | 0.1424               | -2.83        |
| Constant velocity     | 0.1207                           | 0.6215                        | 1.489                | -975.22      |

Table 16: **Recorded DROID: primary external-camera-1 forecasting.** Train-standardized feature MSE at 1, 3 and 5 action blocks (5, 15 and 25 native transitions); lower is better. Learned entries show three-seed mean  $\pm$  sample SD; support-only baselines have no training-seed SD. Gain@5 is  $100(\text{Framewise} - \text{method})/\text{Framewise}$ , a point reduction, not percentage points. **Bold green** marks a positive point reduction, not significance. At five blocks, ours minus the named reference in MSE with 95% paired session/seed bootstrap intervals is Framewise:  $-0.0002808$   $[-0.001038, 0.000413]$  (includes zero); Constant dynamics:  $-0.0002289$   $[-0.0009871, 0.0004498]$  (includes zero). Intervals are unadjusted for multiple comparisons; zero-crossing intervals are inconclusive.

| Method                | External camera 1    |                                 | External camera 2               |                      |
|-----------------------|----------------------|---------------------------------|---------------------------------|----------------------|
|                       | 5 blocks             | 10 blocks                       | 5 blocks                        | 10 blocks            |
| Framewise             | $0.1385 \pm 0.00015$ | $0.1899 \pm 0.00085$            | $0.1543 \pm 8.9 \times 10^{-5}$ | $0.22 \pm 0.00066$   |
| Constant dynamics     | $0.1384 \pm 0.00016$ | $0.1898 \pm 0.00089$            | $0.1543 \pm 9.9 \times 10^{-5}$ | $0.2199 \pm 0.00066$ |
| <b>ShiftWM (ours)</b> | $0.1382 \pm 0.00011$ | $0.1935 \pm 0.004$              | $0.1542 \pm 0.00032$            | $0.2227 \pm 0.0031$  |
| Action-free           | $0.1389 \pm 0.00013$ | $0.1895 \pm 3.9 \times 10^{-5}$ | $0.1544 \pm 0.00016$            | $0.2188 \pm 0.00055$ |
| Persistence           | 0.1424               | 0.1937                          | 0.1574                          | 0.2214               |
| Constant velocity     | 1.489                | 5.114                           | 1.636                           | 5.637                |

Table 17: **Separate recorded-video populations.** Final-horizon standardized MSE; lower is better. External-camera-1/five-block is primary; camera 2 measures natural camera transfer, and ten blocks measure horizon extrapolation. The same session-disjoint test recordings are considered, but complete-window eligibility can differ by horizon. Three-seed means and sample SD are shown for learned methods. These feature forecasts do not measure closed-loop robot success. Raw MSE, cosine error, all paired intervals and reversed-action diagnostics are retained in the source-linked results report.

#### D.1 VALIDATION DEVELOPMENT: DIAGNOSING AND CALIBRATING EXCESS MOTION

After completing the registered study, we investigated overfitting using training and validation recordings only. Stronger command substitutions clarify the reversal diagnostic: replacing future commands with a sequence from another recording session increases the selected ShiftWM model’s ten-block validation error by 5.17%, averaged across three seeds, whereas the action-free control is unchanged. These are sensitivity probes against the original recorded future, not outcomes of executed counterfactual actions. Low-future-motion validation windows suffer relative to persistence, while high-motion windows benefit; these target-defined strata are descriptive and cannot determine an online adaptation rule.

We therefore evaluate a simple, matched residual-calibration control. Let  $r_h = \hat{z}_h - z_0$  denote the predicted displacement from the final observed support feature and  $d_h = z_h - z_0$  its recorded target, both in the frozen training-standardized coordinates. For each selected checkpoint, we fit a single scalar on training recordings:

$$\alpha = \text{clip}_{[0,1]} \frac{\mathbb{E}_{\text{train}} \langle r_h, d_h \rangle}{\mathbb{E}_{\text{train}} \|r_h\|_2^2}, \quad \tilde{z}_h = z_0 + \alpha r_h. \quad (5)$$

We set  $\alpha = 0$  for zero predicted-displacement energy. The expectation averages all ten horizons and feature dimensions, then windows within each episode and equally across eligible training episodes.

| Method                | Validation: 5 blocks |            |              | Validation: 10 blocks |            |              |
|-----------------------|----------------------|------------|--------------|-----------------------|------------|--------------|
|                       | Original             | Calibrated | Gain (%)     | Original              | Calibrated | Gain (%)     |
| Framewise             | 0.159071             | 0.158942   | <b>+0.08</b> | 0.216903              | 0.216206   | <b>+0.32</b> |
| Constant dynamics     | 0.158998             | 0.158880   | <b>+0.07</b> | 0.216746              | 0.216099   | <b>+0.30</b> |
| <b>ShiftWM (ours)</b> | 0.158398             | 0.157704   | <b>+0.44</b> | 0.219507              | 0.215874   | <b>+1.66</b> |
| Action-free           | 0.159766             | 0.159718   | <b>+0.03</b> | 0.217435              | 0.217144   | <b>+0.13</b> |

Table 18: **Validation development: matched residual calibration, not a new test.** Train-standardized feature MSE, lower is better; all four methods and three seeds are included. One scalar per frozen best checkpoint is fitted on 830 eligible training episodes (7,721 windows), with no validation targets used for fitting. Each Gain column is the reduction from that row’s own uncalibrated model. **Bold green** marks positive point reductions, not significance. Against also-calibrated Framewise, ours improves 0.78% at five blocks and 0.15% at ten; the paired MSE differences with 95% session/seed bootstrap intervals are  $-0.001238$   $[-0.002173, -0.000470]$  and  $-0.000332$   $[-0.001395, +0.001022]$ , respectively. The latter includes zero. These are exploratory validation intervals, unadjusted for multiple comparisons; the original held-out results remain unchanged.

The base recursive trajectory is unchanged; contraction is applied to its output. All four methods and three seeds receive the same fitting procedure on 830 eligible training episodes and 7,721 windows; no new neural weights are trained. The selected base checkpoints, data and scientific dependencies remain frozen, and all twelve calibration configurations pass exact offline CPU reload checks.

Table 18 reports the complete validation comparison. Calibration reduces our ten-block error by 1.66% relative to its own uncalibrated model. Against equally calibrated Framewise, the reductions are 0.78% at five blocks and 0.15% at ten; the latter paired interval includes zero. Both horizons use 141 eligible validation episodes from 59 sessions, with 1,772 and 1,631 windows, respectively. This standard least-squares control is development evidence, not a new methodological contribution or a replacement for the original test results. The separately frozen evaluation on untouched recording sessions is reported in Section E; it does not replace the original test results or change the development status of this validation comparison.

## E CONFIRMATORY FORECASTING ON FRESH RECORDING SESSIONS

**An untouched recording population.** After the training/validation calibration study, we registered a deterministic selection of twelve additional official DROID shards, excluding all original twenty-four shards. The new download contains 556 episodes in 293 site/date sessions. Before decoding images or running any model, we excluded every episode whose session appeared anywhere in the original train, validation or test splits: 491 episodes were removed solely for session overlap. The remaining 65 episodes come from 52 sessions absent from all 433 original sessions, with no original episode-ID or serialized-record overlap. We did not resample, enlarge this selection, or filter on outcomes, motion or appearance. Session separation does not establish distinct scenes or objects.

**A frozen comparison.** Before decoding these observations, we separately froze the evaluation code, all twelve selected checkpoints, original training normalization, encoder, twelve train-fitted scalars, metrics and comparisons. We evaluated all four methods, three training seeds and both original and calibrated variants on each of two cameras and two horizons: 96 complete evaluations. Every method receives the same calibration procedure from Equation 5; the scalar is never fitted on fresh observations. No new neural weights are trained. The original causal three-frame support, recorded command blocks, feature extraction and stride-five windowing remain unchanged. Future observations are scoring targets only. All 65 recordings supply five-block windows. One recording lacks enough frames for ten blocks; it remains in the manifest and contributes no fabricated query.

**Prespecified primary endpoint.** The sole primary comparison is calibrated ShiftWM (ours) versus equally calibrated Framewise on camera 1 at the *fifth* predicted frame, measured in original training-standardized feature MSE. This is an endpoint error, not the average across five pre-

| Method                                   | External camera 1 |                 | External camera 2 |                 |
|------------------------------------------|-------------------|-----------------|-------------------|-----------------|
|                                          | 5 blocks          | 10 blocks       | 5 blocks          | 10 blocks       |
| <i>Original selected checkpoints</i>     |                   |                 |                   |                 |
| Framewise                                | 0.148953          | 0.208396        | 0.185354          | 0.220646        |
| Constant dynamics                        | 0.148876          | 0.208241        | 0.185292          | 0.220554        |
| <b>ShiftWM (ours)</b>                    | <b>0.148514</b>   | <b>0.210635</b> | <b>0.185000</b>   | <b>0.222992</b> |
| Action-free                              | 0.149175          | 0.207603        | 0.185950          | 0.220577        |
| <i>Train-fitted residual calibration</i> |                   |                 |                   |                 |
| Framewise                                | 0.148757          | 0.207555        | 0.185222          | 0.219935        |
| Constant dynamics                        | 0.148695          | 0.207461        | 0.185169          | 0.219888        |
| <b>ShiftWM (ours)</b>                    | <b>0.147654</b>   | <b>0.206815</b> | <b>0.184255</b>   | <b>0.219325</b> |
| Action-free                              | 0.149096          | 0.207252        | 0.185900          | 0.220275        |
| Persistence                              | 0.151979          | 0.210400        | 0.190841          | 0.226667        |
| Constant velocity                        | 1.661833          | 6.011535        | 1.753509          | 6.012756        |

Table 19: **Fresh real-DROID sessions: complete original and calibrated comparison.** Final-horizon training-standardized feature MSE ( $\downarrow$ ), averaged equally over episodes and three training seeds. Every method uses its original validation-selected checkpoint; the second block applies its own scalar fitted on original training data only. The two support-only baselines are unchanged across variants. Five-block evaluation uses all 65 episodes from 52 new site/date sessions (866 windows); ten-block evaluation uses 64 eligible episodes from the same 52 sessions (801 windows). The one short episode remains in the audited manifest. Calibrated camera-1/five-block is the prespecified primary population; other populations are separate secondary analyses. These are feature forecasts of actual recorded observations, not physical control outcomes.

dicted frames. Table 19 reports all original and calibrated methods; Table 20 preserves their paired gains and uncertainty. Calibrated ShiftWM obtains 0.147654 versus 0.148757 for calibrated Framewise, a **0.742%** reduction. The paired MSE difference is  $-0.001103$ , with 95% interval  $[-0.002145, -0.000290]$ . Thus the registered primary comparison supports a modest improvement on these fresh sessions. It does not establish a large performance margin or state-of-the-art superiority.

**Secondary results and retained regressions.** Camera-2/five-block transfer yields a 0.522% point reduction against calibrated Framewise, with a descriptive interval below zero. Calibrated ten-block gains are 0.356% on camera 1 and 0.277% on camera 2; both intervals include zero. The original uncalibrated model is worse than Framewise at ten blocks on both cameras: 0.210635 versus 0.208396, and 0.222992 versus 0.220646, respectively. These are relative error increases of 1.075% and 1.063%, and are retained rather than replaced by the calibrated outcomes. Confidence intervals use 10,000 paired session-cluster draws crossed with training-seed resampling, after averaging windows within each episode and weighting eligible episodes equally. Only the stated primary comparison is confirmatory; secondary intervals have no multiplicity adjustment.

**Scope of the evidence.** This separate test supports the reproducibility of a small short-horizon forecasting gain. Residual contraction is a standard least-squares control, not a novel architectural or theoretical contribution. Long-horizon superiority over learned controls remains inconclusive. All input, checkpoint and result hashes were checked, and endpoint arithmetic was independently recomputed from every recorded episode. The evidence concerns observational forecasts of real robot videos; it does not measure closed-loop physical control.

**What changes in individual recordings.** Figure 9B connects recorded inputs to measured forecast gains after the matched calibration in Equation 5. It shows the largest episode improvement, the median episode and the largest regression, alongside the complete distribution: 46 of 65 episodes improve and 19 regress. Episode-average gains need not hold in every window; the fixed first window of the median episode is one such counterexample. The images are recorded observations, while prediction quality is measured in the frozen feature space. Panel A retains the corresponding

| Population                               | Gain (%) $\uparrow$ | Paired MSE difference [95% CI]   |
|------------------------------------------|---------------------|----------------------------------|
| <i>Original selected checkpoints</i>     |                     |                                  |
| Camera 1, 5 blocks                       | <b>+0.294</b>       | -0.000438 [-0.001373, +0.000366] |
| Camera 1, 10 blocks                      | -1.075              | +0.002240 [-0.000480, +0.005574] |
| Camera 2, 5 blocks                       | <b>+0.191</b>       | -0.000353 [-0.001285, +0.000483] |
| Camera 2, 10 blocks                      | -1.063              | +0.002346 [-0.000464, +0.005829] |
| <i>Train-fitted residual calibration</i> |                     |                                  |
| <b>Camera 1, 5 blocks (primary)</b>      | <b>+0.742</b>       | -0.001103 [-0.002145, -0.000290] |
| Camera 1, 10 blocks                      | <b>+0.356</b>       | -0.000740 [-0.002391, +0.000907] |
| Camera 2, 5 blocks                       | <b>+0.522</b>       | -0.000967 [-0.001971, -0.000174] |
| Camera 2, 10 blocks                      | <b>+0.277</b>       | -0.000609 [-0.002464, +0.001283] |

Table 20: **Fresh-session gains and uncertainty versus matched Framewise.** ShiftWM (ours) is compared with Framewise within the same original or calibrated variant. Gains are relative error reductions, not percentage points; **bold green** denotes a positive point estimate, not statistical significance. Intervals use 10,000 paired session-cluster bootstrap draws crossed with training-seed resampling. Only the calibrated camera-1/five-block comparison is confirmatory. Its interval excludes zero; both calibrated ten-block intervals cross zero. All secondary intervals are descriptive and unadjusted for multiplicity. The uncalibrated ten-block regressions are retained explicitly.

original-test comparison, with its distinct population and episode-mean curves. Panel B uses fixed first-window curves; the two studies are not pooled.



Figure 9: **Matched recordings in two separate DROID studies of the historical context model (ours).** Each band retains the previously selected largest-gain, median and largest-regression episodes; selection is outcome-conditioned. Photos are complete recorded input/target frames 10/35 from the fixed first window, not predicted RGB. Gain is matched Framewise minus context-model standardized feature MSE; all MSE gains, curves, ranked marks and intervals are multiplied by 1,000. E/W denote episode/first-window h5 gains. Three-seed mean curves use all episode windows at h1/3/5 in (A), and the first window at h1–5 after matched train-fitted calibration in (B). \*The fresh median episode improves while its shown window regresses. Each action block contains five recorded commands; connecting lines guide the eye. Ranked strips retain all 132/65 episodes. Population intervals are paired 95% session/seed bootstraps; percentages compare population mean errors. Complete absolute method curves remain in the source pack; population values are in Tables 16 and 19.

## F MATCHED GENERALIZATION DEVELOPMENT ON RECORDED VIDEO

This registered development study investigates the early validation optimum in the original recorded-video campaign. Three fixed interventions change either the learning rate (Slow:  $3 \times 10^{-5}$ ), weight decay (Decay: 0.1), or the predictor/context capacity (Compact: width 96, depth 2, context dimensions 16/64). Each arm trains Framewise, Constant dynamics, ShiftWM (ours), and Action-free at three seeds for all 30 epochs. The Compact intervention changes these capacity settings jointly and does not isolate their separate effects.

All inputs, normalization and model selection use the original training and validation partitions. Neither the original test nor the separately acquired fresh holdout selects these revisions. Each checkpoint uses the common all-five-query validation objective. Subsequent five- and ten-block forecasts are summarized by averaging windows within episodes and then equally averaging episodes and seeds. Development improvements, if present, require another independent evaluation for a new confirmatory claim; ordinary optimization and capacity controls are not presented as additional method novelty.

Table 21 reports every matched arm and method. Tables 22 and 23 provide the session/seed paired intervals, including regressions and intervals crossing zero. Table 24 records the selected epochs, parameter counts and measured invocation times. All 36 selected packages pass strict weights-only CPU reload parity on original-validation causal inputs. Figure 10a summarizes the six paired endpoint contrasts alongside the separate horizon-control study in panels b–c.

Historical context model (ours) · DROID validation



**Figure 10: Optimization, capacity and horizon controls for the historical context model (ours), on original DROID validation.** Context denotes the factorized two-context model; Constant denotes Constant dynamics. (a) Context minus matched Framewise endpoint MSE under Slow, Decay and Compact interventions, at five or ten action blocks. All 36 runs train 30 epochs and use the common five-query selection objective; Compact changes predictor and context capacity jointly. (b) Ten-query-trained minus original five-query-trained mean MSE on identical ten-query-eligible windows, for every method. The separated final row retains the unfavorable matched-five-query Context secondary result ( $-0.116\%$  relative gain). Training and selection horizons change together. (c) Context minus equally ten-query-trained Framewise on standard populations. Mean averages query errors; end denotes the endpoint. All 12 horizon-study runs train 30 epochs; every selected checkpoint is epoch 1. Bars are paired 95% session/seed bootstrap intervals (10,000 draws); open marks include zero. Axes use different limits but the same MSE unit. Windows are averaged within 141 episodes from 59 sessions and then three seeds equally. Standard h5 uses 1,772 windows; h10 and matched h5 prefixes use 1,631. All four-method absolute errors remain in Tables 21, 26 and 27; all 20 registered horizon contrasts remain in the tables and companion ledger. These exploratory, multiplicity-unadjusted comparisons provide no independent-test claim.

| Arm     | Method                | MSE@5 ↓ | MSE@10 ↓ | Gain@5 (%)   | Gain@10 (%)  |
|---------|-----------------------|---------|----------|--------------|--------------|
| Slow    | Framewise             | 0.15848 | 0.21626  | reference    | reference    |
| Slow    | Constant dynamics     | 0.15841 | 0.21605  | <b>+0.04</b> | <b>+0.09</b> |
| Slow    | <b>ShiftWM (ours)</b> | 0.15802 | 0.21556  | <b>+0.29</b> | <b>+0.32</b> |
| Slow    | Action-free           | 0.15902 | 0.21564  | -0.34        | <b>+0.28</b> |
| Decay   | Framewise             | 0.15907 | 0.21689  | reference    | reference    |
| Decay   | Constant dynamics     | 0.15899 | 0.21673  | <b>+0.05</b> | <b>+0.07</b> |
| Decay   | <b>ShiftWM (ours)</b> | 0.15839 | 0.21948  | <b>+0.43</b> | -1.20        |
| Decay   | Action-free           | 0.15976 | 0.21742  | -0.44        | -0.25        |
| Compact | Framewise             | 0.15964 | 0.21925  | reference    | reference    |
| Compact | Constant dynamics     | 0.15963 | 0.21923  | <b>+0.01</b> | <b>+0.01</b> |
| Compact | <b>ShiftWM (ours)</b> | 0.15888 | 0.22054  | <b>+0.48</b> | -0.59        |
| Compact | Action-free           | 0.16019 | 0.21877  | -0.34        | <b>+0.22</b> |

Table 21: **Complete matched generalization development study.** All 36 runs completed 30 epochs on original training data. Entries are three-seed, equal-episode validation means; MSE uses fixed train-standardized feature coordinates. Five blocks is the primary development horizon; ten blocks is extrapolation. Slow reduces learning rate, Decay increases weight decay, and Compact jointly reduces predictor/context capacity. Gains compare each method to Framewise within the same arm. **Bold green** identifies positive point reductions, not statistical significance. Every arm and method is retained; these are validation results, not an additional held-out test.

| Arm     | Method                | $\Delta$ MSE | Gain (%)     | Paired 95% interval    |
|---------|-----------------------|--------------|--------------|------------------------|
| Slow    | Constant dynamics     | -0.000065    | <b>+0.04</b> | [-0.000111, -0.000015] |
| Slow    | <b>ShiftWM (ours)</b> | -0.000458    | <b>+0.29</b> | [-0.000671, -0.000283] |
| Slow    | Action-free           | +0.000539    | -0.34        | [-0.000062, +0.001162] |
| Decay   | Constant dynamics     | -0.000073    | <b>+0.05</b> | [-0.000105, -0.000045] |
| Decay   | <b>ShiftWM (ours)</b> | -0.000676    | <b>+0.43</b> | [-0.001535, -0.000053] |
| Decay   | Action-free           | +0.000696    | -0.44        | [+0.000208, +0.001192] |
| Compact | Constant dynamics     | -0.000017    | <b>+0.01</b> | [-0.000072, +0.000039] |
| Compact | <b>ShiftWM (ours)</b> | -0.000759    | <b>+0.48</b> | [-0.001353, -0.000278] |
| Compact | Action-free           | +0.000543    | -0.34        | [+0.000029, +0.001149] |

Table 22: **Matched validation uncertainty at 5 blocks.** Differences are method minus Framewise within each arm, in standardized feature MSE; negative favors the named method. Relative gains are  $100(\text{Framewise} - \text{method})/\text{Framewise}$ . Intervals jointly resample recording sessions and the three training seeds with 10,000 draws. These exploratory development intervals are unadjusted for multiple comparisons; intervals crossing zero do not establish a difference. Positive point gains are colored independently of interval significance.

| Arm     | Method                | $\Delta$ MSE | Gain (%)     | Paired 95% interval    |
|---------|-----------------------|--------------|--------------|------------------------|
| Slow    | Constant dynamics     | -0.000201    | <b>+0.09</b> | [-0.000322, -0.000086] |
| Slow    | <b>ShiftWM (ours)</b> | -0.000698    | <b>+0.32</b> | [-0.001144, -0.000339] |
| Slow    | Action-free           | -0.000610    | <b>+0.28</b> | [-0.002089, +0.000888] |
| Decay   | Constant dynamics     | -0.000157    | <b>+0.07</b> | [-0.000241, -0.000079] |
| Decay   | <b>ShiftWM (ours)</b> | +0.002593    | -1.20        | [-0.000748, +0.005568] |
| Decay   | Action-free           | +0.000532    | -0.25        | [-0.000728, +0.001798] |
| Compact | Constant dynamics     | -0.000017    | <b>+0.01</b> | [-0.000194, +0.000153] |
| Compact | <b>ShiftWM (ours)</b> | +0.001291    | -0.59        | [-0.000945, +0.003425] |
| Compact | Action-free           | -0.000483    | <b>+0.22</b> | [-0.002460, +0.001390] |

Table 23: **Matched validation uncertainty at 10 blocks.** Differences are method minus Framewise within each arm, in standardized feature MSE; negative favors the named method. Relative gains are  $100(\text{Framewise} - \text{method})/\text{Framewise}$ . Intervals jointly resample recording sessions and the three training seeds with 10,000 draws. These exploratory development intervals are unadjusted for multiple comparisons; intervals crossing zero do not establish a difference. Positive point gains are colored independently of interval significance.

| Arm     | Method                | Best epochs | Total (M) | Trainable (M) | Seconds/run    |
|---------|-----------------------|-------------|-----------|---------------|----------------|
| Slow    | Framewise             | 3/3/3       | 3.763     | 3.469         | 151 [149, 219] |
| Slow    | Constant dynamics     | 3/3/3       | 3.763     | 3.714         | 159 [159, 234] |
| Slow    | <b>ShiftWM (ours)</b> | 3/3/3       | 3.763     | 3.714         | 161 [161, 236] |
| Slow    | Action-free           | 3/3/3       | 3.763     | 3.312         | 141 [140, 213] |
| Decay   | Framewise             | 1/1/1       | 3.763     | 3.469         | 148 [146, 218] |
| Decay   | Constant dynamics     | 1/1/1       | 3.763     | 3.714         | 158 [156, 238] |
| Decay   | <b>ShiftWM (ours)</b> | 2/2/1       | 3.763     | 3.714         | 161 [158, 240] |
| Decay   | Action-free           | 1/1/1       | 3.763     | 3.312         | 140 [138, 213] |
| Compact | Framewise             | 2/2/2       | 0.841     | 0.763         | 100 [99, 159]  |
| Compact | Constant dynamics     | 2/2/2       | 0.841     | 0.828         | 110 [110, 175] |
| Compact | <b>ShiftWM (ours)</b> | 2/3/3       | 0.841     | 0.828         | 112 [111, 173] |
| Compact | Action-free           | 2/1/2       | 0.841     | 0.722         | 93 [93, 149]   |

Table 24: **Reusable checkpoints and computation.** Best epochs are ordered by seeds 0/1/2 and selected by the shared all-five-query validation objective after full 30-epoch training. Parameter counts include frozen parameters; trainable counts exclude inactive modules. Seconds/run is the median [minimum, maximum] successful runner invocation, including training, validation and CPU reload verification; prior interrupted work, if any, is excluded. Different GPU partitions are used, so these times are operational evidence, not a hardware-matched speed comparison. All 36 selected packages pass strict CPU reload parity.

## G TESTING CAUSAL SUPPORT RELIABILITY

We tested whether errors within a longer observed prefix predict when the factorized forecast should be trusted over Framewise. This is a calibration and ensembling diagnostic, not a new architectural contribution. The protocol, code, donor checkpoints and decision rule were frozen before execution; only original training and validation recordings were used.

**Causal prefix backtesting.** Both frozen donors received their respective original training-fitted residual calibrations. Within thirteen observed frames, each donor forecast the last ten from the first three and recorded commands, without incorporating any of those ten targets into its context or rollout. Let  $d$  denote the factorized-minus- Framewise forecast and  $e$  the observed-minus-Framewise residual, both in fixed training-standardized coordinates. The candidate sets

$$\alpha_s = \text{clip}_{[0,1]} \frac{\langle e, d \rangle_s + \lambda \alpha_0}{\langle d, d \rangle_s + \lambda}, \quad (6)$$

where each inner product averages prefix horizons and feature dimensions. The zero-denominator fallback is  $\alpha_0$ . The prior and shrinkage scale were fitted using original training only; three-fold training-session cross-validation selected from four fixed shrinkage values. All three seeds selected  $\alpha_0 = 1$  and  $\kappa = 100$ , where  $\lambda = \kappa$  times training-average donor-disagreement energy. The base donors had already trained on those sessions, so this procedure cross-validates only gate fitting, not end-to-end model training.

For the actual query, both donors restarted from the final three observed prefix frames. Their forecasts were blended with fixed  $\alpha_s$ , using recorded future commands and no future images. All gate packages were locked before decoding validation features. The longer-prefix population contains 758 training episodes in 295 sessions and 132 validation episodes in 57 sessions. Respectively 93 and 11 recordings were too short for the common 23-frame window and remain in the audit. This observation budget differs from the main three-frame-support experiment. The candidate requires both donors and two additional prefix rollouts; it is not a single-model improvement.

**Failed promotion and complete controls.** Table 25 retains all eleven arms, including simple averaging, training-fitted constant blends, unshrunk, one-step and session-shuffled gates. The candidate’s h5 reduction versus the global prior is 0.007%, with paired MSE difference  $-0.00001134$  and 95% interval  $[-0.00002378, +0.00000256]$ . Its h10 reduction is 0.022%, with interval  $[-0.00008776, +0.00000306]$  for the paired MSE difference. It fails the prespecified 1% gain, interval and all-seed conditions. Ordinary equal averaging has lower mean error than the adaptive candidate; this does not establish that prefix reliability transfers to the next query. We therefore do not promote this candidate or test it on the previously inspected holdouts. Intervals resample sessions and training seeds and remain exploratory. Table errors use FP64 sufficient-statistic arithmetic on frozen FP32 donor predictions. The exported FP32 wrapper has exact reload parity; this is not a claim of bitwise parity with FP64 score arithmetic.

| Method                                 | Endpoint MSE ↓ |          | Gain vs. global prior ↑ |               |
|----------------------------------------|----------------|----------|-------------------------|---------------|
|                                        | h5             | h10      | h5 (%)                  | h10 (%)       |
| Framewise, calibrated                  | 0.152472       | 0.213742 | -0.464                  | -0.055        |
| ShiftWM (ours), calibrated             | 0.151768       | 0.213624 | +0.000                  | +0.000        |
| Equal blend ( $\alpha = 0.5$ )         | 0.151513       | 0.211785 | <b>+0.168</b>           | <b>+0.861</b> |
| Global-prior blend (reference)         | 0.151768       | 0.213624 | +0.000                  | +0.000        |
| Train-query constant blend             | 0.151768       | 0.213624 | +0.000                  | +0.000        |
| Unshrunk support gate                  | 0.151606       | 0.212266 | <b>+0.107</b>           | <b>+0.636</b> |
| <b>Shrunk support gate (candidate)</b> | 0.151757       | 0.213577 | <b>+0.007</b>           | <b>+0.022</b> |
| One-step support gate                  | 0.151761       | 0.213595 | <b>+0.005</b>           | <b>+0.013</b> |
| Shuffled support gate                  | 0.151761       | 0.213590 | <b>+0.004</b>           | <b>+0.016</b> |
| Persistence                            | 0.155705       | 0.215908 | -2.594                  | -1.069        |
| Constant velocity                      | 1.916075       | 6.879813 | -1162.502               | -3120.525     |

Table 25: **A support-reliability candidate that did not pass its development gate.** Original camera-1 validation only: 132 episodes, 57 sessions and 1,351 common windows, equally weighted over episodes and three seeds. All arms have thirteen observed prefix frames and identical ten-step query windows, whereas the main forecasting protocol uses three support frames. Gains are relative to the *global-prior blend*, which equals calibrated ShiftWM because all training-fitted priors are one. **Bold green** marks positive point estimates, not significance or a passed decision rule. The candidate improves h5 by only 0.007%; its paired difference is  $-0.00001134$ , with 95% interval  $[-0.00002378, +0.00000256]$ . Ordinary equal averaging has the lowest mean errors. The candidate fails the prespecified 1% improvement, interval and all-seed criteria; it is retained as a negative development result. No new neural weights are trained.

## H MATCHED TEN-STEP TRAINING ON RECORDED VIDEO

This registered control tests whether training at the evaluation horizon improves long-horizon prediction. It reuses the unchanged width-192, depth-4 LeWM-based predictor with Framewise, Constant dynamics, ShiftWM (ours), and Action-free variants. All twelve runs train for 30 epochs at seeds 0/1/2, using three observed support frames and ten recursively predicted query targets. The optimizer, training-only normalization, action blocks, camera, training stride, and validation stride match the original five-step campaign.

No query image conditions forecasts or inferred contexts. Model selection uses FP32 standardized MSE averaged over all ten query steps and windows, preserving the original window weighting. An equal-episode metric is logged only as an auxiliary diagnostic. All evaluation tables equally weight episodes and then training seeds, as in the original evaluation. Training and checkpoint-selection horizons change together; this ordinary horizon-mismatch control is not a new architecture or evidence of additional method novelty.

Tables 26 and 27 compare all h10-trained methods on each standard validation population. Table 28 qualifies ours–Framewise differences using paired seed/session intervals. Tables 29 and 30 compare original and new checkpoints on identical h10-eligible query starts, including their five-step prefixes. The primary diagnostic is mean error across all ten query predictions; endpoint and five-step effects are secondary. Positive point gains, regressions and intervals including zero are all retained. These exploratory validation results require independent evaluation before any new confirmatory claim. Neither original nor fresh test payloads enter this study. Figure 10b–c shows the matched horizon and method contrasts, including the unfavorable five-step secondary outcome.

Table 31 documents the selected reusable checkpoints and measured invocation times. The accompanying evidence ledger records every configuration, checkpoint, metric source and independent numerical check.

| Method                | Endpoint MSE@5 | Endpoint MSE@10 | Gain@5 (%)   | Gain@10 (%)  |
|-----------------------|----------------|-----------------|--------------|--------------|
| Framewise             | 0.15866        | 0.21429         | reference    | reference    |
| Constant dynamics     | 0.15863        | 0.21423         | <b>+0.02</b> | <b>+0.03</b> |
| <b>ShiftWM (ours)</b> | 0.15830        | 0.21342         | <b>+0.23</b> | <b>+0.40</b> |
| Action-free           | 0.15936        | 0.21496         | -0.44        | -0.31        |

Table 26: **Matched h10-trained methods: endpoint mse.** Every method completed 30 epochs with width 192, depth 4 and three support frames, at seeds 0/1/2. Values are means across seeds and equally weighted validation episodes; all four methods use identical populations within each horizon. The standard h5 and h10 populations may differ because each requires a complete query window. Gains use Framewise as the reference within the same horizon. **Bold green** marks positive point reductions, not significance. All methods and regressions are retained; these are development results, not a new test evaluation.

| Method                | Mean query MSE@5 | Mean query MSE@10 | Gain@5 (%)   | Gain@10 (%)  |
|-----------------------|------------------|-------------------|--------------|--------------|
| Framewise             | 0.11286          | 0.15261           | reference    | reference    |
| Constant dynamics     | 0.11284          | 0.15258           | <b>+0.01</b> | <b>+0.02</b> |
| <b>ShiftWM (ours)</b> | 0.11271          | 0.15222           | <b>+0.13</b> | <b>+0.25</b> |
| Action-free           | 0.11323          | 0.15322           | -0.33        | -0.40        |

Table 27: **Matched h10-trained methods: mean query mse.** Every method completed 30 epochs with width 192, depth 4 and three support frames, at seeds 0/1/2. Values are means across seeds and equally weighted validation episodes; all four methods use identical populations within each horizon. The standard h5 and h10 populations may differ because each requires a complete query window. Gains use Framewise as the reference within the same horizon. **Bold green** marks positive point reductions, not significance. All methods and regressions are retained; these are development results, not a new test evaluation.

| Scoring     | Ours    | Framewise | Gain (%)     | Paired 95% interval  |
|-------------|---------|-----------|--------------|----------------------|
| Endpoint 5  | 0.15830 | 0.15866   | <b>+0.23</b> | [-0.00052, -0.00022] |
| Mean 1–5    | 0.11271 | 0.11286   | <b>+0.13</b> | [-0.00022, -0.00008] |
| Endpoint 10 | 0.21342 | 0.21429   | <b>+0.40</b> | [-0.00133, -0.00049] |
| Mean 1–10   | 0.15222 | 0.15261   | <b>+0.25</b> | [-0.00059, -0.00022] |

Table 28: **ShiftWM (ours) versus equally h10-trained Framewise.** Intervals are for ours minus Framewise in standardized feature MSE, so negative favors ours. The primary control metric averages all ten query predictions. Endpoint metrics and mean-five errors are secondary. Intervals jointly resample the three training seeds and recording-session clusters (10,000 draws); they are exploratory and unadjusted for multiple comparisons. † marks intervals including zero. Green bold point gains alone do not establish a difference.

| Method                | $h$ | Original h5 | Trained h10 | Gain (%)                 | Paired 95% interval  |
|-----------------------|-----|-------------|-------------|--------------------------|----------------------|
| Framewise             | 5   | 0.10972     | 0.10962     | <b>+0.09<sup>†</sup></b> | [-0.00028, +0.00007] |
| Framewise             | 10  | 0.15346     | 0.15261     | <b>+0.56</b>             | [-0.00135, -0.00039] |
| Constant dynamics     | 5   | 0.10968     | 0.10961     | <b>+0.07<sup>†</sup></b> | [-0.00025, +0.00009] |
| Constant dynamics     | 10  | 0.15338     | 0.15258     | <b>+0.52</b>             | [-0.00130, -0.00034] |
| <b>ShiftWM (ours)</b> | 5   | 0.10938     | 0.10951     | -0.12 <sup>†</sup>       | [-0.00022, +0.00059] |
| <b>ShiftWM (ours)</b> | 10  | 0.15385     | 0.15222     | <b>+1.06</b>             | [-0.00272, -0.00080] |
| Action-free           | 5   | 0.11011     | 0.11002     | <b>+0.08<sup>†</sup></b> | [-0.00022, +0.00005] |
| Action-free           | 10  | 0.15403     | 0.15322     | <b>+0.52</b>             | [-0.00114, -0.00046] |

Table 29: **Matched training-horizon control: mean errors.** Original h5-trained and new h10-trained checkpoints use the exact same h10-eligible episode/session identities and query starts. The h5 rows use the five-step prefix of those windows; h10 uses all ten. These rows average all query steps. Gains are  $100(\text{original} - \text{h10})/\text{original}$ ; paired intervals are h10 minus original. Both selection objectives retain window weighting but use their respective training horizons. <sup>†</sup> marks intervals including zero; green bold identifies only positive point gains. All four methods are shown, with no independent-test claim.

| Method                | $h$ | Original h5 | Trained h10 | Gain (%)                 | Paired 95% interval  |
|-----------------------|-----|-------------|-------------|--------------------------|----------------------|
| Framewise             | 5   | 0.15671     | 0.15630     | <b>+0.26</b>             | [-0.00081, -0.00002] |
| Framewise             | 10  | 0.21690     | 0.21429     | <b>+1.21</b>             | [-0.00383, -0.00148] |
| Constant dynamics     | 5   | 0.15664     | 0.15627     | <b>+0.23<sup>†</sup></b> | [-0.00076, +0.00002] |
| Constant dynamics     | 10  | 0.21675     | 0.21423     | <b>+1.16</b>             | [-0.00375, -0.00138] |
| <b>ShiftWM (ours)</b> | 5   | 0.15623     | 0.15600     | <b>+0.15<sup>†</sup></b> | [-0.00090, +0.00061] |
| <b>ShiftWM (ours)</b> | 10  | 0.21951     | 0.21342     | <b>+2.77</b>             | [-0.00936, -0.00309] |
| Action-free           | 5   | 0.15749     | 0.15709     | <b>+0.25</b>             | [-0.00067, -0.00010] |
| Action-free           | 10  | 0.21743     | 0.21496     | <b>+1.14</b>             | [-0.00323, -0.00168] |

Table 30: **Matched training-horizon control: endpoint errors.** Original h5-trained and new h10-trained checkpoints use the exact same h10-eligible episode/session identities and query starts. The h5 rows use the five-step prefix of those windows; h10 uses all ten. These rows score the final query step. Gains are  $100(\text{original} - \text{h10})/\text{original}$ ; paired intervals are h10 minus original. Both selection objectives retain window weighting but use their respective training horizons. <sup>†</sup> marks intervals including zero; green bold identifies only positive point gains. All four methods are shown, with no independent-test claim.

| Method                | Best epochs | Total (M) | Trained (M) | Seconds/run    |
|-----------------------|-------------|-----------|-------------|----------------|
| Framewise             | 1/1/1       | 3.763     | 3.469       | 249 [242, 336] |
| Constant dynamics     | 1/1/1       | 3.763     | 3.714       | 262 [256, 368] |
| <b>ShiftWM (ours)</b> | 1/1/1       | 3.763     | 3.714       | 264 [258, 368] |
| Action-free           | 1/1/1       | 3.763     | 3.312       | 234 [229, 321] |

Table 31: **Horizon-ten checkpoints and computation.** Best epochs follow seeds 0/1/2, selected by window-weighted FP32 validation MSE over all ten recursive predictions after complete 30-epoch training. Total parameters include inactive frozen modules; trained parameters exclude them. Seconds/run is the median [minimum, maximum] successful runner invocation, including training, five evaluation passes and CPU reload checks, excluding prior interrupted work. Hardware partitions differ, so invocation times are operational records, not a matched speed comparison. All 12 selected packages pass registered and independently repeated weights-only CPU reload checks.

## I ORIGINAL CONTEXT STUDY: METHOD AND PRIMARY COMPARISONS

This appendix preserves the original context-based protocol and its completed primary planning comparisons. In these original-study tables and figures, ShiftWM denotes Ours-1. The subsequent spatial variants use separate representations and development protocols.

### I.1 PROBLEM AND INFORMATION ACCESS

Let  $x_t$  be hidden physical state,  $a_t$  an executed action,  $p$  a dynamics setting, and  $v$  an observation setting. Within a stationary segment,

$$x_{t+1} = F_p(x_t, a_t, \epsilon_t), \quad o_t = R_v(x_t, \eta_t). \quad (7)$$

The deployed agent receives images and actions, not  $x_t, p, v$ . Training covers selected pairs  $(v, p)$ . Compositional testing holds out a pair while retaining each factor individually in training; extrapolation to new values is a separate evaluation. Appearance transforms preserve the action coordinate system. Camera geometry changes and noninvertible occlusion require different protocols.

An earlier support segment  $S$  contains  $H$  observations and  $H - 1$  actions; later query images  $Q$  are disjoint from support. At deployment, support is real past history, and the controller receives a goal image. Imagined futures never become evidence for context inference. Complete trajectory families, including appearance-paired renders, stay within one data split.

**Training supervision and fairness.** A canonical render depicts the *same physical state* under a reference appearance; changing dynamics requires a new rollout. All trained controls receive the same trajectories, canonical targets, training schedule, and validation selection. ShiftWM additionally uses pairing identities for context consistency. Shared versus unpaired contexts tests architecture; unpaired contexts versus ShiftWM tests the extra consistency supervision. A comparison against the frozen model alone cannot attribute a gain to factorization.

### I.2 SHIFTWM: OBSERVATION AND DYNAMICS CONTEXTS

The visual encoder and projector  $E_0$  are frozen. We train context networks, adapters, the action encoder, predictor, and prediction projector offline. Weights remain fixed online. This historical model combines shared observation calibration, corrected-transition dynamics inference and paired supervision; the LeWM backbone and CEM planner are reused. Equations 8–11 specify these computations. Appendix K gives dimensions, initialization and parameter counts. The current spatial decoder is defined separately in Section 3 and its consolidated architecture in Figure 2.

#### I.2.1 CONTEXT INFERENCE AND PREDICTION

For  $u_t = E_0(o_t) \in \mathbb{R}^d$ , the observation pathway computes

$$c_o = C_o(u_S), \quad z_t = A_o(u_t, c_o). \quad (8)$$

The residual adapter  $A_o$  calibrates shifted features to reference coordinates. Dynamics context uses chronological corrected support and executed actions:

$$c_d = C_d((z_i, a_i, z_{i+1})_{i \in S}), \\ \tilde{a}_t = A_d(E_a(a_t), c_d), \quad \hat{z}_{t+1} = P_\theta(z_{t-L+1:t}, \tilde{a}_{t-L+1:t}). \quad (9)$$

Here  $A_d$  is an action-embedding adapter and  $L$  is predictor history. Correcting support before dynamics inference is an inductive bias, not proof of independence; uninformative actions can leave dynamics ambiguous. Contexts are inferred once per replan and held fixed over candidate rollouts.

#### I.2.2 LEARNING IN FIXED COORDINATES

For a canonical training render  $o_t^* = R_{v_0}(x_t)$ , the target is  $z_t^* = \text{sg}(E_0(o_t^*))$ . The prediction loss  $\mathcal{L}_{\text{pred}}$  is teacher-forced one-step query MSE, and  $\mathcal{L}_{\text{align}}$  is corrected-feature MSE to the same frozen reference. Appearance variants of the same trajectory encourage equal dynamics contexts through



Figure 11: **Crossed appearance and dynamics in the historical simulation study.** Full recorded simulator frames illustrate PushT and Reacher. The four PushT views apply the exact RGB transforms to one state; physical settings change damping or arm and finger density together. Training and ordinary validation use seven pairs;  $(v_1, p_1)$  is development and  $(v_2, p_2)$  is held-out composition. Three additional pairs use new factor values; blank cells are unevaluated. Independent test trajectories cover all nine in-range pairs, including familiar combinations. Trajectory seeds are split before paired rendering; factor IDs are unavailable to the deployed model. This is the historical context-model protocol.

$\mathcal{L}_d$ . Different batch examples sharing an appearance label encourage equal observation contexts through  $\mathcal{L}_o$ . The objective is

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda_{\text{align}} \mathcal{L}_{\text{align}} + \lambda_d \mathcal{L}_d + \lambda_o \mathcal{L}_o, \quad (10)$$

with weights  $(1, 0.1, 0.01)$ . Unpaired contexts disable the two consistency terms. Targets have fixed coordinates, but the contexts themselves may collapse or rescale with their adapters. Constant-context and shuffled-context controls are therefore required before interpreting the contexts as useful factor-specific information. Appendix K.2 states the exact index sets and batch-pairing rules, including the mismatch between one-step training and recursive evaluation.

### I.2.3 PLANNING

The goal embedding is  $z_g = A_o(E_0(o_g), c_o)$ , assuming the goal shares the current observation setting. A common CEM/MPC solver minimizes

$$J(a_{t:t+K-1}) = \text{mse}(\hat{z}_{t+K}, z_g), \quad (11)$$

executes the first action chunk, and replans from new real observations. All methods share candidate count, iterations, action bounds, horizon, and paid interaction budget. Different-view goals and online gradient adaptation are outside the completed primary comparison.

## I.3 EXPERIMENTS

**Tasks and composition.** We use visual PushT and Reacher in `stable-worldmodel`. Appearance factors are canonical, warm, and cool RGB transforms. PushT varies damping; Reacher varies arm and finger density together. Seven factor pairs train the model,  $(v_1, p_1)$  is development, and  $(v_2, p_2)$  is held-out composition (Figure 11). Separate extrapolation tests introduce dim appearance and a new physical value. Each environment has 256 training, 32 validation, 32 development, and 64 test initial-state seeds. Test trajectories are independent and cover all nine in-range pairs; extrapolation covers three additional pairs. Appendix J gives exact values, episode counts, and action interfaces.

**Compared methods.** *Frozen LeWM* is the released pretrained reference. *Framewise calibration*, *Shared context*, and *Unpaired contexts* are our matched controls on that same backbone, not independent reproductions of published methods (Table 32). Framewise calibration tests whether per-image correction suffices. Shared context tests whether one inferred context is sufficient. Unpaired contexts retains the two-branch architecture while removing context-consistency losses. *ShiftWM (ours)* combines separate contexts with those paired losses. Every trained variant has three seeds

Table 32: **Methods in the primary comparison.** Frozen LeWM is the released reference; the other rows are matched in-house implementations on its backbone. Every learned visual calibration path uses canonical alignment. Only ShiftWM uses paired context consistency. None uses online gradients.

| Method                | Visual calibration | Context  | Consistency |
|-----------------------|--------------------|----------|-------------|
| Frozen LeWM           | None               | None     | None        |
| Framewise calibration | Per frame          | None     | None        |
| Shared context        | Contextual         | Shared   | None        |
| Unpaired contexts     | Contextual         | Separate | None        |
| <b>ShiftWM (ours)</b> | Contextual         | Separate | Paired      |

| Method                          | PushT               |                    | Reacher             |                    |
|---------------------------------|---------------------|--------------------|---------------------|--------------------|
|                                 | Held-out $\uparrow$ | Extrap. $\uparrow$ | Held-out $\uparrow$ | Extrap. $\uparrow$ |
| Frozen LeWM                     | 4.7                 | 7.3                | 35.9                | 17.7               |
| Framewise calibration           | $3.6 \pm 0.9$       | $5.0 \pm 0.8$      | $42.7 \pm 3.3$      | $21.2 \pm 5.9$     |
| Shared context                  | $2.1 \pm 0.9$       | $7.6 \pm 1.1$      | $36.5 \pm 3.9$      | $16.8 \pm 3.3$     |
| Unpaired contexts               | $2.6 \pm 0.9$       | $5.2 \pm 1.4$      | $34.9 \pm 9.0$      | $20.8 \pm 2.6$     |
| <b>ShiftWM (ours)</b>           | $2.6 \pm 0.9$       | $6.8 \pm 0.5$      | $40.6 \pm 3.1$      | $19.6 \pm 1.3$     |
| $\Delta$ vs Shared context (pp) | <b>+0.52</b>        | -0.87              | <b>+4.17</b>        | <b>+2.78</b>       |
| 95% paired CI                   | [-1.04, +2.08]      | [-2.95, +1.22]     | [-4.17, +12.50]     | [-1.91, +7.64]     |

Table 33: **Main comparison: closed-loop planning success (%)**. Higher is better. Trained entries require all three seeds and show mean  $\pm$  sample SD; Frozen LeWM uses one released checkpoint per environment. The final rows give ShiftWM minus Shared context in percentage points and conditional 95% paired task-cluster intervals, not SD. **Bold green** marks positive mean differences, not statistical significance. Dashes denote incomplete evidence, not zero success. Held-out is  $(v_2, p_2)$ ; extrapolation averages three prespecified conditions. All methods share the paid support and planning budget. Framewise calibration, Shared context and Unpaired contexts are in-house controls on the same LeWM backbone. Full metrics and the unaligned diagnostic appear in Table 39.

and 30 completed epochs; ordinary-validation prediction error selects checkpoints. The *Unaligned predictor* retains mismatched observed/goal coordinates and is reported only as a diagnostic in Appendix S.

**Metrics and evaluation budget.** Closed-loop success is primary. The main comparison separates the held-out pair from extrapolation; canonical and all-condition results remain in the appendix. Planning uses 300 CEM candidates, 30 iterations, 30 elites, a five-group horizon, and five native controls per group. The 50-native-step budget includes ten support-acquisition steps. Methods use matched initial-state/goal identities and common support. Raw success includes support-only successes; policy-eligible success excludes them. Table entries require all three trained seeds and report mean  $\pm$  sample standard deviation, while paired task-cluster intervals appear separately in Appendix T. Forecast MSE@5 uses five recursive transitions with recorded actions in immutable reference coordinates. Hyperparameters are fixed without test-based selection; elapsed inference time is not compared without dedicated hardware measurements.

#### I.4 RESULTS AND ANALYSIS

All 30 training runs, main forecast evaluations, and 64 prescribed checkpoint-split planning evaluations are complete. Table 33 is the main closed-loop comparison; every trained-method entry now includes all three prescribed seeds and passes source and protocol validation. Full condition results, eligibility counts, and exact forecast values are in Appendix S.

**Does composition improve planning?** The complete matched comparisons show mixed effects. Relative to Framewise calibration, held-out raw-success differences are  $-1.04$  percentage points

on PushT (conditional 95% interval  $[-3.65, 1.56]$ ) and  $-2.08$  on Reacher ( $[-11.46, 7.29]$ ). On extrapolation, PushT improves by  $+1.74$  points ( $[0.00, 3.65]$ ), while Reacher declines by  $-1.56$  ( $[-6.25, 3.13]$ ). These intervals condition on the three observed training seeds.

Positive held-out point estimates occur against Shared context ( $+0.52$  points on PushT and  $+4.17$  on Reacher) and against Unpaired contexts on Reacher ( $+5.73$ ); PushT ties Unpaired contexts. Across the 12 raw-success contrasts in Table 41, six means are positive, five negative, and one zero. Every conditional 95% interval includes zero; the PushT extrapolation interval against Framewise touches it exactly. The intervals are not multiplicity-adjusted and do not capture uncertainty over the population of training runs. The completed evidence therefore does not establish consistent planning superiority or isolate a benefit from context pairing. Policy-eligible comparisons, with support-only successes excluded, are reported alongside the raw contrasts. All prescribed three-seed planning comparisons with Framewise calibration and Unpaired contexts are available in Table 33, with paired uncertainty reported in Table 41. Differences only against Frozen LeWM would conflate the proposed structure with additional training.

**Where ShiftWM improves forecasting.** Against the fixed Framewise calibration reference, ShiftWM achieves **26.78% lower held-out Reacher MSE@5** and 3.68% lower PushT MSE@5, using ratios of three-seed means (Figure 12). The closest aligned held-out comparator is Unpaired contexts: adding paired consistency reduces its error by 9.21% on Reacher and 2.30% on PushT. These are relative forecast-error reductions, not changes in planning success.

Relative MSE reduction (%)  $\uparrow$

| Reference             | PushT        |              | Reacher       |         |
|-----------------------|--------------|--------------|---------------|---------|
|                       | Held-out     | Extrap.      | Held-out      | Extrap. |
| Framewise calibration | <b>+3.68</b> | <b>+4.20</b> | <b>+26.78</b> | -4.15   |
| Unpaired contexts     | <b>+2.30</b> | <b>+0.70</b> | <b>+9.21</b>  | -0.35   |

The summary defines reduction as  $100(\text{baseline} - \text{ours}) / \text{baseline}$ , so positive means lower error and negative means a regression. **Bold green** marks positive mean reductions, not statistical significance. On extrapolation, PushT improves over Framewise calibration by 4.20%, while Reacher worsens by 4.15%; Frozen LeWM has the lowest Reacher extrapolation error among the aligned models. The unaligned diagnostic has lower held-out PushT error than ShiftWM and remains visible in the full results. Thus, the held-out gains do not establish uniform superiority. Unpaired contexts also improves on Shared context in both held-out environments, separating architectural changes from pairing. Mean ordering and three-seed dispersion do not identify the factors encoded by either context or constitute a significance test.



Figure 12: **Forecasting on held-out composition and extrapolation.** Terminal latent MSE after five recursive transitions (25 native actions) using recorded future actions; lower is better. Points and ranges show three-training-seed means and sample standard deviations, not confidence intervals. Frozen LeWM has one checkpoint and no training-seed error bar. Extrapolation averages the three prespecified new-value conditions. Horizontal axes are logarithmic with panel-specific limits; latent coordinates differ by environment. Gain annotations use signed error change relative to Framewise calibration: negative means lower MSE. These ratios of three-seed means are point summaries, not significance estimates. Forecast accuracy alone does not establish planning success.

**Why better forecasts need not yield better control.** In the completed seed-0 development comparison, ShiftWM and Shared context have equal success rates; Framewise calibration has the highest Reacher point estimate. A subsequent goal-only intervention keeps the ShiftWM predictor fixed but substitutes framewise goal correction. Eligible Reacher success increases from 6/30 to 12/30, compared with 11/30 for the complete framewise controller; PushT falls from 1/31 to 0/31 (Appendix Q). This post-hoc intervention motivates separate checks of prediction and goal alignment, not a general repair. CEM also selects its own candidate actions, so mean forecast error on recorded trajectories does not control error on those selected candidates. Mechanism tests, the completed dynamics-residual revision, and the completed controlled rollout-objective study are described in Appendices P, U, and V. Matched observed rollouts in Appendix O show successes, failures, and goals reached during support before planning.

## J COMPLETE EXPERIMENTAL PROTOCOL

**Environments and splits.** The core tasks are visual PushT and Reacher in *stable-worldmodel*. They have different control geometries but share infrastructure. Initial observation factors are canonical, warm, and cool pointwise RGB transforms. PushT changes space damping (0, 0.15, 0.65); Reacher changes arm and finger density together (1000, 650, 1350 in simulator units). We train on seven combinations, reserve  $(v_1, p_1)$  for development, and hold out  $(v_2, p_2)$  for the main compositional test (Figure 11). Test trajectories are independent across the full grid. Additional dim and physical conditions test extrapolation separately: dim appearance  $v_3$ , PushT damping 0.95, and Reacher density 1500, evaluated at  $(v_3, p_0)$ ,  $(v_0, p_3)$ ,  $(v_3, p_3)$ . Each

environment has 256 training, 32 validation, 32 development, and 64 test initial-state seeds. Three physical conditions in the first three splits and four in test produce 1,216 physical episodes per environment, each with 64 grouped transitions. Appearance versions reuse those physical trajectories. The full test planning grid has  $64 \times 9 = 576$  tasks per checkpoint; the separate extrapolation grid has  $64 \times 3 = 192$ . The all-condition test average includes familiar factor pairs and is not a wholly unseen-composition score. This controlled extension is not a reproduction of the original paper’s success table.

## J.1 PLANNING, ELIGIBILITY, AND RESOURCE MEASUREMENT

**Primary outcomes and uncertainty.** Closed-loop success under each environment’s declared criterion is primary. We also report steps to goal and worst-condition success under fixed budgets. Methods use matched initial-state/goal IDs. Core comparisons target three training seeds; actual seed and episode counts accompany each result. Episode-level outcomes support paired comparisons; uncertainty over tasks and variability over training seeds are reported separately. Correlated frames are not independent trials. Failed runs and timeouts remain in the record, with a stated policy for their inclusion in denominators. The main planner uses 300 candidates, 30 CEM iterations, and 30 elites, matching the released LeWM CEM sampling configuration. This budget is locked for all model comparisons before full-test learned-policy planning. It evaluates 9,000 candidate sequences per replan, versus 640 in the earlier development-only setting; this  $14.0625\times$  count ratio is not a measured wall-time ratio. Both settings use five grouped actions of prediction horizon, and one group executed per replan. A group contains five native controls. The 50-step native budget includes ten steps acquiring initial support. CEM optimizes in the checkpoint’s training action z-score coordinates; it converts candidates to native controls and clips only physical bounds  $[-1, 1]$ . It does not clip z-scores to that range. PushT uses the relative-displacement interface from the official training archive; the earlier absolute-control corpus is quarantined. Success during support is recorded separately; policy-eligible success excludes tasks already solved by that common history. Final-state success is also retained. Per-replan latency divides actual solve time by actual replanning calls, excluding nonexistent calls on already-solved tasks. These are configuration facts, not measured costs. Main efficiency results additionally require dedicated execution with no concurrent training on the measured GPU; shared-GPU development timing is retained in raw records but excluded from efficiency tables.

## J.2 ADDITIONAL DIAGNOSTICS AND TASK SCOPE

**Diagnostics.** Completed forecast evaluations measure recursive error in frozen reference coordinates. Action shuffling, context replacement/removal, and support-length dependence are additional mechanism tests; their outcomes are not yet reported. A low one-step error can arise from ignoring actions on predictable trajectories; action interventions and planning test that failure. Prediction diagnostics therefore include persistence, zero-action rollouts, and the change induced by actions. Zero-action predictions are compared with recorded-action targets only as a sensitivity check, not as errors against observed counterfactual trajectories. PushT additionally reports a prespecified subset requiring block translation beyond 20 pixels or rotation beyond  $\pi/9$ , so tasks solved by moving the agent alone do not hide manipulation failures. Both the full set and this subset remain visible. Within-episode switches are a secondary evaluation after stationary conditions. Test combinations are never used for hyperparameter selection.

Action-conditioned phantom endoscopy is an optional offline extension, outside the minimum control claim. Forecasting alone cannot demonstrate navigation or clinical benefit. Static diagnostic images do not provide the executed actions required by this study.

## K IMPLEMENTATION AND TRAINING DETAILS

### K.1 ARCHITECTURE AND PARAMETER ACCOUNTING

The initial implementation freezes a pretrained LeWM visual encoder and projector, denoted  $E_0$ . Context networks, adapters, action encoder, predictor, and prediction projector are trained. Full visual fine-tuning is a distinct configuration. The frozen path permits feature caching without changing

inputs or targets. Equations 8–11 specify this historical computation. The upstream architecture contains 18,034,478 parameters. The ShiftWM *active inference path* uses 18,319,598, with 12,025,454 trainable offline. Shared context has 12,025,046 trainable parameters. The current loaded model object contains 24,873,110 parameters including the duplicated frozen reference encoder and inactive ablation modules; the current package serializes that full state. Active-path counts are not claims about package size or runtime memory, which must be measured on the actual released artifact.

For  $d = 192$  and  $H = L = 3$ , both contexts have width 32; the dynamics network uses a 128-unit GRU. The observation network pools support-feature mean and population variance. Both adapters use bounded-scale residual FiLM. Inference holds contexts fixed within a search, so imagined futures cannot modify the information used to infer dynamics.

Framewise calibration uses the per-image residual  $u + \text{MLP}(\text{LN}(u))$  with dimensions  $192 \rightarrow 128 \rightarrow 192$  and an initially zero output layer. It corrects history and goal images identically, receives no temporal context, and uses the prediction and canonical-alignment losses. It adds 49,856 active parameters; its active path has 18,084,334 parameters, of which 11,790,190 are trained offline. Shared context matches the context-network parameter count approximately; architecture attribution must still account for computation differences.

## K.2 EXACT LOSSES AND TEMPORAL INDEXING

Let  $o_t^* = R_{v_0}(x_t)$  be a paired canonical training render and  $z_t^* = \text{sg}(E_0(o_t^*))$  its frozen target. For a sequence indexed  $0, \dots, T-1$ , context uses only indices  $0, \dots, H-1$ . Define  $\mathcal{T}_{\text{pred}} = \{H, \dots, T-2\}$  and  $\mathcal{T}_{\text{align}} = \{H, \dots, T-1\}$ . The implemented query losses are

$$\mathcal{L}_{\text{pred}} = \frac{1}{|\mathcal{T}_{\text{pred}}|} \sum_{t \in \mathcal{T}_{\text{pred}}} \text{mse}(\hat{z}_{t+1}, z_{t+1}^*), \quad (12)$$

$$\mathcal{L}_{\text{align}} = \frac{1}{|\mathcal{T}_{\text{align}}|} \sum_{t \in \mathcal{T}_{\text{align}}} \text{mse}(z_t, z_t^*). \quad (13)$$

Here  $\text{mse}(u, v) = \|u - v\|_2^2 / \dim(u)$  averages over feature dimensions, matching the implementation and stated loss weights. Training uses teacher-forced one-step query prediction. With  $T = 8$  and  $H = 3$ , context sees frames 0–2, alignment supervises frames 3–7, and prediction supervises targets 4–7 from observed history windows. Frame 3 is not a prediction target in this campaign. Evaluation instead starts immediately after support and recursively predicts frames 3–7. This boundary and teacher-forcing mismatch can affect planning even when validation prediction error is low; the completed runs are not multi-step rollout training. Frozen targets prevent an encoder from lowering the reported loss solely by shrinking or changing its coordinates; they do not guarantee that those coordinates retain every necessary physical quantity.

Appearance variants  $S^{(v,p)}$  and  $S^{(v',p')}$  of the same trajectory give

$$\mathcal{L}_d = \mathbb{E} \text{mse}(C_d(S^{(v,p)}), C_d(S^{(v',p')})), \quad (14)$$

where  $C_d(S)$  abbreviates the corrected-support pathway. Distinct batch examples with the same observation label give the regularizer

$$\mathcal{L}_o = \mathbb{E}_{v,p,p',S,S'} \text{mse}(C_o(S^{(v,p)}), C_o(S^{(v',p')})). \quad (15)$$

The implementation averages over off-diagonal batch pairs with matching appearance IDs; it does not require their dynamics values or trajectory IDs to differ. Grouping establishes a shared transformation, not a shared state. The term does not force state latents of different trajectories to agree. The objective is

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda_{\text{align}} \mathcal{L}_{\text{align}} + \lambda_d \mathcal{L}_d + \lambda_o \mathcal{L}_o. \quad (16)$$

The initial configuration uses  $\lambda_{\text{align}} = 1$ ,  $\lambda_d = 0.1$ , and  $\lambda_o = 0.01$ ; unpaired controls disable both context consistency terms. No additional SIGReg term is used because the target coordinates are frozen. These choices are configuration facts, not tuned results. Raw context consistency also has a scale ambiguity: shrinking a context and increasing downstream sensitivity can lower its penalty without improving factor separation. Consistency can collapse contexts to constants. Constant-context, context-shuffle, and removed-pathway controls therefore test whether both contexts carry useful information. Context swapping across trajectories is not part of the initial implementation: it requires valid matching targets.

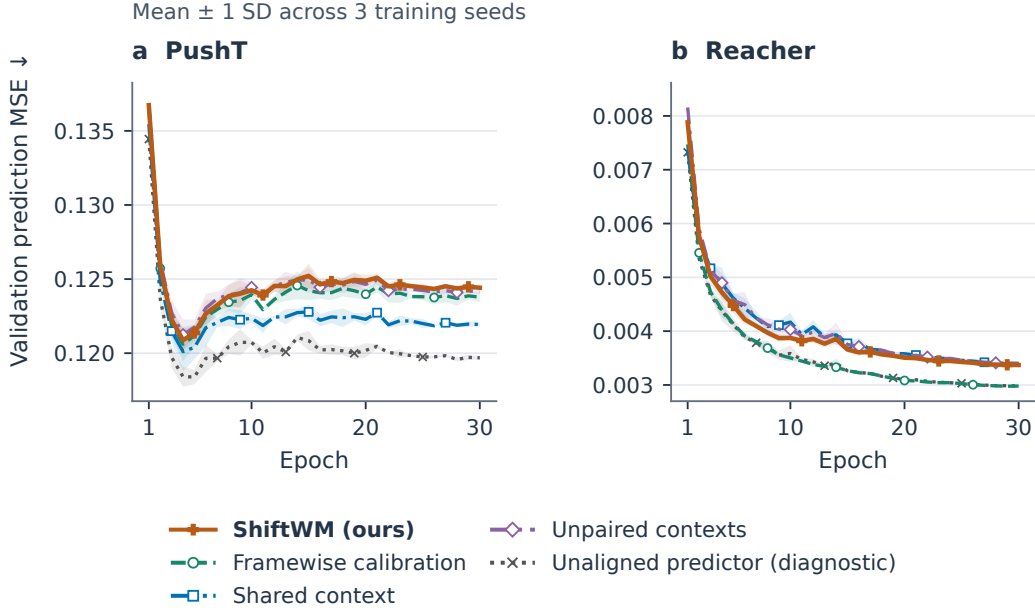


Figure 13: **Validation prediction loss over the completed training campaign.** Curves and shaded bands show means and sample standard deviations over three training seeds at common logged epochs. Colors and line styles identify model variants. No future epochs are extrapolated, and bands are not confidence intervals. The two panels have separate vertical scales. These are validation records, not test-set or planning-performance results.

### K.3 TRAINING AND CHECKPOINT SELECTION

The reported campaign uses the fixed encoder and feature-cache path, with RGB inputs in  $[0, 1]$ , ImageNet channel normalization applied once, and antialiased bilinear resizing to the released encoder input size. The visual projector and prediction projector retain pretrained BatchNorm running statistics. The prediction projector’s affine parameters remain trainable; freezing running statistics prevents query-frame batch statistics from entering support features and permits batch-size-one planning. The full campaign trains five variants (Unaligned predictor, Framewise calibration, Shared context, Unpaired contexts, and ShiftWM), three seeds, and two environments: 30 runs of 30 epochs, batch size 128, stride one, and initial learning rate  $5 \times 10^{-5}$ . All 30 configured runs have completed their 30 epochs; the generated ledger below records the completion evidence. AdamW uses weight decay  $10^{-3}$ , gradient clipping at norm 1, and a cosine schedule ending at  $10^{-6}$ . Training and validation use BF16 autocast when supported; evaluation uses FP32. Checkpoints are selected by ordinary-validation prediction MSE, not last-epoch loss or total loss, whose alignment and consistency terms differ across controls. Full training completion is distinct from selecting an early validation-best checkpoint. Main planning evaluation remains separate.

Both training and CEM retain the released-data action normalization. In the new PushT training corpus, the per-coordinate native-action standard deviations are approximately 0.504 and 0.504, versus 0.208 and 0.207 in the released archive (245,760 collected training actions). This distribution difference is retained in the protocol, not retrospectively removed by renormalizing the completed campaign. Reacher’s corresponding collected and released deviations are both approximately 0.577. These statistics describe action distributions; they do not establish that normalization explains the planning failures.

**Completion ledger.** The configured campaign contains 30 runs. At this artifact snapshot, 30 have recorded at least one complete epoch and 30 have a verified completion summary and final checkpoint. Missing epoch logs do not establish whether a scheduler job is queued or running.

Table 34: Fields required for every released checkpoint.

| Field             | Required record                                                      |
|-------------------|----------------------------------------------------------------------|
| Base architecture | LeWM source commit, checkpoint revision and weight hash              |
| Visual input      | Shape, channel order, normalization, transformation version          |
| Action interface  | Meaning, units, bounds, statistics, frame skip                       |
| Context           | $H$ , dimensions, pooling/sequence modules, reset behavior           |
| Prediction        | $L$ , predictor configuration, recursive loss horizon, goal protocol |
| Loss              | All weights; appearance pairing and batch grouping rules             |
| Optimization      | Optimizer, schedule, batch, steps, precision, trained modules        |
| Data              | Episodes, transitions, seeds, generation and factor/split manifests  |
| Planning          | Candidates, iterations, elite fraction, horizon, action chunk        |
| Resources         | GPU, peak memory, training hours, prediction/planning latency        |

Training curves show observed validation prediction loss only; they do not establish closed-loop performance.

## L INFORMATION ACCESS AND CORRECTNESS

The support must precede query targets. No paired variant of an underlying trajectory can cross a split. Batch grouping can use factor IDs but model inputs cannot. Goal images convey target states, not hidden future trajectories. Future realized actions and rewards cannot enter context used to select an earlier action.

Checks must establish that saved models reload identically in evaluation mode; removing adapters recovers the specified base computation; prediction responds to action changes; support cannot replace query inputs; cached features match online features under identical transforms; checkpoint resumes preserve optimizer/scheduler and random state; and paired rendering does not mutate physical state. These are correctness requirements, not replacements for training and closed-loop evaluation.

**What is not identifiable.** For  $x_{t+1} = x_t + pa_t$ ,  $o_t = vx_t$ , and  $x_0 = 0$ , observations depend on the product  $vp$ . For every  $q > 0$ ,  $(v, p)$  and  $(qv, p/q)$  generate the same observations under every action sequence. Separate physical interpretation therefore needs additional assumptions or information. Our contexts are functional prediction variables, not identified friction or camera parameters. This elementary example is a limitation, not a theoretical novelty claim.

## M STATISTICS AND REPORTING

Every episode records training seed, condition, initial-state and goal IDs, success, length, and failure reason. A paired bootstrap resamples independent task IDs while retaining method pairing; training-seed dispersion is reported separately. The final interval level, calculation, resample count, and actual sample counts accompany each completed figure. Frame count must not inflate the number of independent trials.

The reported campaign fixes support length, loss weights, and planning settings in its versioned configurations; no hyperparameter search over these choices is claimed. Ordinary validation prediction error selects checkpoints. Development diagnostics inform separately labeled follow-up hypotheses. Framework calibration and Unpaired contexts were added after an implementation/validation audit, before the main learned-policy test campaign; the original runs remain unchanged. Test evaluation follows the frozen protocol. A documented implementation error can trigger a complete matched rerun; disappointing performance cannot be relabeled as an installation failure. Aborted and numerically unstable runs remain in the audit trail.

## N DEVELOPMENT COMPARISONS AND BUDGET HISTORY

### N.1 MATCHED OFFICIAL-BUDGET COMPARISON

**Paired-support audit.** PushT: the same 32 seed/trajectory identities, goal indices, support-success masks, eligibility masks and post-support distances agree across 4 completed modes. Reacher: the same 32 seed/trajectory identities, goal indices, support-success masks, eligibility masks and post-support distances agree across 4 completed modes. The comparison currently contains 8/8 completed model/environment records. Unfinished records do not contribute outcomes. Comparative conclusions remain pending; the main multi-seed test and its reporting gate are unchanged.

The seed-0 rates do not establish a multi-seed ordering. In Reacher, ShiftWM and Shared context each solve six eligible tasks but share only one success; equal aggregate rates therefore do not imply equal decisions or solved tasks.

### N.2 EARLIER DEVELOPMENT-BUDGET CHECK

The first complete development check used the validation-selected checkpoint from a completed 30-epoch seed-zero factorized run in each environment, on all 32 prespecified development tasks at  $(v_1, p_1)$ , with 128 candidates, five iterations, and 16 elites. PushT achieved 1/32 total successes; 1 occurred during common support, leaving 0/31 policy-eligible successes. Reacher achieved 11/32 total successes, including 2 during support, with 9/30 policy-eligible successes. These are development diagnostics from one trained seed, not full-test results or evidence of factorization benefit.

| Environment | Model                 | Raw success | During support | Policy-eligible |
|-------------|-----------------------|-------------|----------------|-----------------|
| PushT       | Frozen LeWM           | 2/32        | 1/32           | 1/31            |
| PushT       | Shared context        | 2/32        | 1/32           | 1/31            |
| PushT       | ShiftWM (ours)        | 2/32        | 1/32           | 1/31            |
| PushT       | Framewise calibration | 2/32        | 1/32           | 1/31            |
| Reacher     | Frozen LeWM           | 5/32        | 2/32           | 3/30            |
| Reacher     | Shared context        | 8/32        | 2/32           | 6/30            |
| Reacher     | ShiftWM (ours)        | 8/32        | 2/32           | 6/30            |
| Reacher     | Framewise calibration | 13/32       | 2/32           | 11/30           |

Table 35: **Official-budget development comparison, separate from the main test.** Each completed row uses all 32 prespecified development tasks at  $(v_1, p_1)$ , with 300 CEM candidates, 30 iterations and 30 elites. Raw and support rates use 32 tasks; policy-eligible rates exclude tasks already solved during common support. Trained models use seed 0; frozen LeWM has no new training seed. Dashes denote unfinished rows. These counts do not establish a multi-seed or factorization benefit; changing the CEM budget is not itself evidence of improvement.

Baseline comparisons and a matched rerun at the main sampling budget are needed to interpret them. The official sampling-budget discrepancy was corrected before learned-policy full-test planning began. Increasing the search budget is not itself evidence that planning improves. Shared-GPU timings from this check are excluded from efficiency claims. This historical note records the pre-comparison budget discrepancy. The matched official-budget comparison above has since completed; the earlier check is retained without merging its outcomes into the main comparison.

## O MATCHED QUALITATIVE EXECUTIONS

Figures 16–17 show actual simulator observations from the completed seed-zero development evaluations at  $(v_1, p_1)$ . We select the lexicographically first task in each outcome stratum relative to Framewise calibration: ShiftWM-only success, Framewise-only success, and joint failure. The fourth example is joint success for Reacher; PushT has no eligible joint success, so we explicitly show a support-only success instead. Selection is by this stated rule, not visual quality; these outcome-stratified examples do not estimate success frequency. The accompanying local gallery contains all 64 tasks and adds Shared context.

Both methods receive pixel-identical initial observations and available goals, with the same planner seed and budget: 300 candidates, 30 CEM iterations, 30 elites, horizon five action blocks, and at most 50 native calls including support. The first saved frame follows support acquisition; subsequent frames follow executed blocks of up to five actions. Labels reconstruct exact native times, including early stopping. Repeated terminal observations do not imply additional interactions. The world model predicts latents; none of the displayed images is a decoded prediction, interpolated state, or generated illustration.

### O.1 FROM THE INFERENCE PATH TO MEASURED EXECUTION

Figure 14 joins the historical inference paths with matched execution diagnostics. Framewise uses a temporal, action-conditioned predictor with an image residual applied independently to each frame. ShiftWM infers an observation context from support statistics and a dynamics context from corrected transitions and executed actions. The former calibrates history and goal; the latter modulates action embeddings. Goals and future actions never enter context inference. The methods share the frozen encoder and predictor architecture but have separately trained predictor weights. Deployment weights stay fixed, while contexts update after observed action blocks and remain fixed inside each CEM search. This historical context model is distinct from the spatial model in Figure 2.

Panels B and C examine the first positive discordant case in each environment under the declared lexicographic rule. We replay saved actions for *all thirteen* discordant tasks across three methods, including nine Framewise-only successes. Replay matches every archived goal and observed frame, support and terminal errors, and stopping behavior. Reconstructed physical states are measurement targets, never policy inputs or newly selected actions.

Each task panel separates both conditions required for success. Green regions show the original tolerances, and traces stop at the actual native call. The lower plots test both checkpoints on identical three-frame histories, two past action blocks and the next executed five-action block, against the same canonical next-observation feature. Canonical images are evaluator-only targets. After common support, the policies visit different states, so comparisons are paired within each realized transition. Partial terminal blocks have no full-block measurement; they are omitted, never plotted as zero.



**Figure 14: Historical computation and matched-transition diagnostics.** (A) Methods share a frozen encoder, temporal architecture and CEM procedure, with separately trained predictors. Support mean/variance infer  $c_o$  for history/goal calibration; corrected transitions and executed past actions infer  $c_d$  for action-embedding FiLM. Goals and future actions never enter context inference. Deployment weights stay fixed; contexts are recomputed after observations, then fixed within each search. (B–C) The original first positive discordant case per task retains its goal and two observed call-20 states. Reacher uses the original common crop, marked on the full goal overview; dashed goal contours are display-only. PushT’s green T is unscored. Both task tolerances must hold simultaneously: combined pusher/block position error < 20 native-512 pixels and wrapped angle <  $\pi/9$ , or both unwrapped Reacher joint errors < 0.05 rad. Traces stop at actual calls 21/23 versus 50; green regions show the separate tolerances. Lower plots pair the two predictors on identical three-frame histories and executed five-action blocks, against the same frozen canonical next-feature target. Circles/solid lines denote ShiftWM (ours); squares/dashes denote Framewise. Grouping names the behavior policy, whose visited states differ after support. Incomplete terminal blocks are omitted, not zero. ShiftWM has lower prediction error on 0/5 PushT and 3/5 Reacher pairs; across all four positive cases it is lower on 6/20 prediction and 0/20 calibration diagnostics. These outcome-conditioned, dependent observations neither establish component causality nor estimate planning success frequency.

In PushT, ShiftWM meets both tolerances at call 21 while Framewise fails at call 50, despite ShiftWM having higher one-block prediction error on every illustrated transition. In Reacher, ShiftWM reaches both joint tolerances at call 23. Prediction MSE is 42.5% and 37.0% lower along

its two full transitions, but lower on only one of three transitions along Framewise’s execution. These diagnostics do not recover CEM candidate rankings or establish which component caused success. Across the four positive cases, ShiftWM has lower next-observation prediction MSE on 6 of 20 matched transitions and lower observed-history calibration MSE on 0 of 20. These outcome-selected, temporally dependent transitions are descriptive counts, not independent benchmark wins or significance estimates. Success therefore cannot be equated with improvement in every intermediate model measurement.

## O.2 ADDITIONAL GOAL-REACHING EXAMPLES

Figure 15 completes the positive cases from Figure 14: together they show all four policy-eligible development tasks on which the historical ShiftWM (ours) context model succeeds and Framewise fails. Shared context fails on each. The full eligible comparison has one versus one discordant success on PushT and three versus eight on Reacher; these selected cases explain outcomes, not average superiority. Counterexamples follow, and the gallery retains every task.

Identical per-case crops and display-only goal outlines expose the remaining joint mismatch. Call-20 frames align observation times, while endpoints retain the actual stopping times. The endpoint joint-error plots show why visual closeness or a single L2 threshold does not determine success: both unwrapped joint errors must be below 0.05 radians. These observations support a behavioral explanation without attributing success to a particular context or candidate ranking. Appendix V separately examines objective and context interventions.

### Two additional Reacher successes · historical context model

#### (a) Episode 2031008



#### (b) Episode 2031011

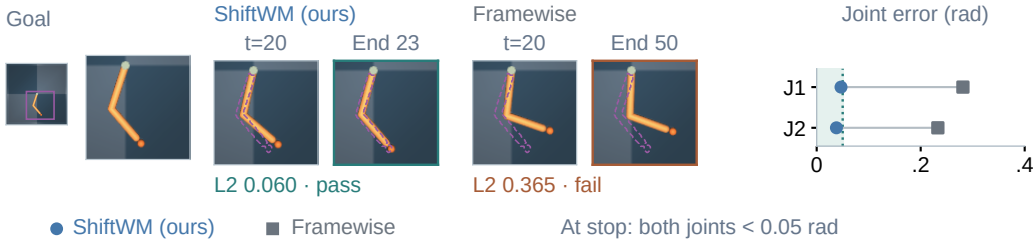


Figure 15: **Two additional positive Reacher cases of the historical context model (ours).** Both methods share the goal, ten support actions and a 50-action allowance. Recorded images compare native call 20 and each method’s actual endpoint. Every detail uses the original common per-case crop, located in the full-goal overview; dashed goal outlines are display-only. Dots show raw unwrapped absolute joint errors at each endpoint. Success requires *both* errors strictly below 0.05 radians, so the L2 distance 0.060 in (b) can pass. These are the two remaining ShiftWM-only cases beyond episodes 2031024 and 2031004 shown in the technical comparisons. Shared context fails in all four. Selection is outcome-conditioned: the full eligible development comparison has one versus one discordant success on PushT and three versus eight on Reacher.

## O.3 CONTRASTING SUCCESSES, FAILURES, AND SUPPORT-ONLY OUTCOMES

The compact outcome plates use identical goal and support observations within each task. Their terminal frames retain each method’s actual stopping time. The PushT support-only case makes

clear why success before any CEM plan is excluded from eligible policy success. Full intermediate rollouts and Shared-context comparisons remain in the archived gallery.



**Figure 16: Matched PushT outcomes, including success before planning.** Recorded simulator observations compare the historical ShiftWM (ours) context model with Framewise, at training seed 0 on development ( $v_1, p_1$ ). These are the original first-lexicographic cases from four outcome strata, not success-frequency estimates. Each card shares pixel-identical goal and support observations across methods. Success/failure labels give the actual native stopping call; endpoints are not synchronized. The scored goal is the gray block and blue pusher in the goal image. The persistent green T is an unscored reset marker. Block and pusher errors use native  $512 \times 512$  coordinates despite the  $224 \times 224$  recorded images. Success requires their combined position-error norm below 20 pixels and wrapped block-angle error below  $\pi/9$ . In (b), 1.2 pixels of block error does not compensate for 429.0 pixels of pusher error. In (d), every method reaches the goal during shared support at call 2; the support and terminal image is identical and shown once. No CEM plan executes. This raw success is excluded from eligible policy success and is not evidence of planner competence; no eligible joint-success PushT case exists in this population. The allowance is 50 native calls including support. All Shared-context outcomes and intermediate observations remain in the complete archived gallery; no displayed image is an RGB prediction.



Figure 17: **Matched Reacher outcomes across all four preselected strata.** Observed executions compare the historical ShiftWM (ours) context model with Framewise calibration. Each card shares a goal and the identical observation after ten support calls; both arrays match exactly across methods. Endpoint labels give success/failure and the actual native stopping call, not a synchronized endpoint. Joint-angle L2 distance (radians) is diagnostic: success requires every unwrapped joint error to be below 0.05 radians. These are the original first-lexicographic development cases at training seed 0 with a 50-call allowance; strata illustrate outcomes, not their frequency. Shared context and all intermediate observations remain in the complete archived gallery, including Shared context failure on the case where both displayed methods succeed. Images are recorded simulator observations, not RGB predictions or current spatial-decoder outputs.

## P MECHANISM TESTS AND ADDITIONAL BASELINES

The remaining tests distinguish useful context information from a constant vector, and observation correction from predictor fine-tuning alone. Consistency penalties have a scale ambiguity: context magnitudes can shrink while downstream sensitivity grows. Small penalties or context variance do not establish factor separation. Interventions should measure adapter outputs, prediction changes, and planning with the same available history. Constant-context, context-shuffle, removed-pathway, and support-length controls remain incomplete. Within-episode switching is a secondary study.

Observation-only and dynamics-only *context* ablations still train the shared predictor; they are not encoder-only or dynamics-only gradient adaptation. The official AdaJEPA reproduction is complete under its separate released protocol (Appendix W.3); an input- and budget-matched comparison on our tasks remains unfinished. Matched online gradient updates and oracle contexts also remain planned controls. Their supervision, optimization, data-generation cost, parameter counts, and measured time must accompany performance.

Table 36: **Planned mechanism and resource analysis.** These targeted comparisons are incomplete; dashes are not measured zeros. Completed core forecast comparisons appear separately in Figure 12. Each future row must specify its split, support length, and measured resources.

| Variant                      | Success $\uparrow$ | Error $\downarrow$ | ms/replan | Trainable M |
|------------------------------|--------------------|--------------------|-----------|-------------|
| ShiftWM (ours)               | —                  | —                  | —         | —           |
| Without observation context  | —                  | —                  | —         | —           |
| Without dynamics context     | —                  | —                  | —         | —           |
| Unpaired contexts            | —                  | —                  | —         | —           |
| Shuffled dynamics support    | —                  | —                  | —         | —           |
| Shared context, matched size | —                  | —                  | —         | —           |
| Joint gradient adaptation    | —                  | —                  | —         | —           |

## Q DEVELOPMENT GOAL-CALIBRATION DIAGNOSTIC

After completing the prespecified development comparison, we inspect goal calibration separately from future prediction. Each diagnostic uses the same goal, support history, and support-success exclusion as planning. The corrected shifted goal and the five-block recorded-action prediction are compared with the same immutable canonical goal embedding. Recorded future actions and canonical renders are evaluator-only diagnostic inputs; they are not available to the deployed controller. There are 31 eligible PushT cases and 30 eligible Reacher cases, selected from all 32 development tasks per environment using the common support-success mask. Fresh GPU Reacher support and goal images exactly match their stored canonical counterparts in every eligible case.

For ShiftWM on Reacher, recorded-action prediction MSE is approximately 0.012 while corrected-goal MSE is 0.797. Framewise calibration yields approximately 0.009 and 0.271, respectively. This is evidence of an error mismatch, not proof that goal calibration causes the planning deficit. A separate development intervention substitutes the learned framewise goal correction while holding the ShiftWM predictor, history correction, context inference and planner fixed. This intervention is separate from the primary method comparison.

### Q.1 AN ELEMENTARY COST-ERROR DECOMPOSITION

This calculation explains why prediction and goal calibration require separate checks; it is not a theoretical novelty or an empirical guarantee. For a nonempty finite set of candidate action sequences  $\mathcal{A}$  containing the returned sequence, let  $y(a)$  and  $g$  be canonical terminal and goal latents, and let  $\hat{y}(a), \hat{g}$  be their model estimates. Define  $\|u\|_d = \|u\|_2/\sqrt{d}$ , so squared normalized norm equals the implemented MSE. Suppose, uniformly over  $\mathcal{A}$ ,

$$\|\hat{y}(a) - y(a)\|_d \leq e_f, \quad \|\hat{g} - g\|_d \leq e_g, \quad \|y(a) - g\|_d \leq R.$$

Writing  $c(a) = \|y(a) - g\|_d^2$ ,  $\hat{c}(a) = \|\hat{y}(a) - \hat{g}\|_d^2$  and  $\varepsilon = e_f + e_g$ , expansion of the squared norm gives

$$|\hat{c}(a) - c(a)| \leq 2R\varepsilon + \varepsilon^2 =: B.$$

If a returned action sequence satisfies  $\hat{c}(\hat{a}) \leq \min_{a \in \mathcal{A}} \hat{c}(a) + \eta$ , then

$$c(\hat{a}) - \min_{a \in \mathcal{A}} c(a) \leq 2B + \eta = 4R\varepsilon + 2\varepsilon^2 + \eta.$$

Indeed, for a true minimizer  $a^*$ , apply the cost-error bound twice:  $c(\hat{a}) \leq \hat{c}(\hat{a}) + B \leq \hat{c}(a^*) + \eta + B \leq c(a^*) + 2B + \eta$ . An illustrative scalar counterexample has exact predictions  $y(a) = 0, y(b) = 1$ , true goal  $g = 0$  and estimated goal  $\hat{g} = 1$ : goal error alone reverses the two candidates' ranking. This is a mathematical example, not a measured simulator result. Empirical mean prediction MSE does not provide the assumed uniform norm bound; CEM does not certify  $\eta$ . The statement concerns latent cost on the stated candidate set, not physical success or causal attribution for the observed failures.

Development diagnostic · mean and 95% bootstrap interval



Figure 18: **Prediction and goal calibration are distinct errors.** Posthoc development diagnostics for frozen LeWM and three seed-0 trained models. Both errors are per-dimension MSE in fixed pre-trained coordinates against the same canonical goal; lower is better. Blue circles denote goal calibration; orange diamonds use the five recorded future action blocks. Horizontal axes are logarithmic so small errors remain visible. Error bars are 95% trajectory-seed bootstrap intervals, excluding common support-only successes; they do not measure training-seed variation. Small recorded-action prediction error alone does not establish accurate goal calibration or successful planning.

| Environment | Method                      | Raw   | Support | Eligible |
|-------------|-----------------------------|-------|---------|----------|
| PushT       | ShiftWM (ours)              | 2/32  | 1/32    | 1/31     |
| PushT       | Shared context              | 2/32  | 1/32    | 1/31     |
| PushT       | Framewise calibration       | 2/32  | 1/32    | 1/31     |
| PushT       | Goal-only hybrid (post hoc) | 1/32  | 1/32    | 0/31     |
| Reacher     | ShiftWM (ours)              | 8/32  | 2/32    | 6/30     |
| Reacher     | Shared context              | 8/32  | 2/32    | 6/30     |
| Reacher     | Framewise calibration       | 13/32 | 2/32    | 11/30    |
| Reacher     | Goal-only hybrid (post hoc) | 14/32 | 2/32    | 12/30    |

Table 37: Exploratory development results at training seed 0. The post-hoc hybrid changes only the fixed ShiftWM model’s learned goal calibration. Raw success includes support-only successes; eligible success excludes them. All methods use the same 32 development tasks and 300/30/30 CEM budget. Pending runs are not zero-valued outcomes.

| Environment | Paired comparison              | Wins/losses | $\Delta$ (pp) | 95% interval (pp) |
|-------------|--------------------------------|-------------|---------------|-------------------|
| PushT       | Hybrid — ShiftWM (ours)        | 0/1         | -3.2          | [-9.7, +0.0]      |
| PushT       | Hybrid — Framewise calibration | 0/1         | -3.2          | [-9.7, +0.0]      |
| Reacher     | Hybrid — ShiftWM (ours)        | 9/3         | <b>+20.0</b>  | [+0.0, +40.0]     |
| Reacher     | Hybrid — Framewise calibration | 7/6         | <b>+3.3</b>   | [-20.0, +26.7]    |

Table 38: Paired eligible-success differences for the post-hoc hybrid. Green bold means indicate positive differences against the named comparator, not statistical significance. Wins/losses count discordant task outcomes. Intervals resample paired development trajectories (20,000 draws, seed 1701); they do not measure variation across training seeds and are exploratory, not confirmatory tests.

## R RELEASED-EVALUATOR IMPLEMENTATION CHECK

The unchanged released LeWM evaluation entry point, released checkpoints, and released datasets yield 46/50 PushT successes and 39/50 Reacher successes on the official seed-42 task selections. These runs use a separately installed, API-compatible May-2026 `stable-worldmodel` snapshot. The exact authors’ training runtime was not specified and is not claimed to have been recovered. Both executions completed, and configuration, task identities, checkpoint hashes and all 50 Boolean outcomes were verified from the run records. This check supports the functioning of the released planning pathway on its released data. It does not test equivalence of that pathway to our model wrapper on identical episodes, reproduce a published aggregate score or establish equivalence with our current simulator version. Its expert-data PushT tasks and original 25-step replanning interval differ from the new intervention benchmark and its paid 10-step support phase. These scores are therefore kept separate from every adaptation comparison.

## S FULL CONDITION AND DIAGNOSTIC RESULTS

The aligned methods share corrected history/goal coordinates. The *Unaligned predictor (diagnostic)* instead predicts canonical targets while shifted goals remain uncorrected. This mismatch makes it an unsuitable primary comparator for a factorization claim, although its forecast errors remain informative. Its alignment term is constant with no gradient; training curves compare prediction loss rather than different total losses. Canonical and all-condition averages retain nominal regressions that a held-out-only result could conceal. The all-condition average contains familiar factor pairs and is not a wholly unseen-composition score.

| Method                            | Planning success (%) $\uparrow$ |                       |                       |                       | Feature error $\downarrow$ |
|-----------------------------------|---------------------------------|-----------------------|-----------------------|-----------------------|----------------------------|
|                                   | Canonical                       | Held-out              | Eligible              | Extrap.               | MSE@5                      |
| <b>PushT / LeWM transformer</b>   |                                 |                       |                       |                       |                            |
| Frozen LeWM                       | <b>29.7</b>                     | <b>4.7</b>            | <b>3.2</b>            | 7.3                   | 1.1854                     |
| Framewise calibration             | 10.9 $\pm$ 1.6                  | 3.6 $\pm$ 0.9         | 2.1 $\pm$ 0.9         | 5.0 $\pm$ 0.8         | 0.5713 $\pm$ 0.0155        |
| Shared context                    | 14.6 $\pm$ 3.3                  | 2.1 $\pm$ 0.9         | 0.5 $\pm$ 0.9         | <b>7.6</b> $\pm$ 1.1  | 0.7216 $\pm$ 0.0287        |
| Unpaired contexts                 | 16.1 $\pm$ 5.5                  | 2.6 $\pm$ 0.9         | 1.1 $\pm$ 0.9         | 5.2 $\pm$ 1.4         | 0.5633 $\pm$ 0.0083        |
| ShiftWM (ours)                    | 16.7 $\pm$ 0.9                  | 2.6 $\pm$ 0.9         | 1.1 $\pm$ 0.9         | 6.8 $\pm$ 0.5         | 0.5503 $\pm$ 0.0099        |
| Unaligned diagnostic              | 13.0 $\pm$ 3.9                  | 2.1 $\pm$ 0.9         | 0.5 $\pm$ 0.9         | 6.1 $\pm$ 1.1         | <b>0.5168</b> $\pm$ 0.0067 |
| Ours — Framewise / gain           | <b>+5.73</b>                    | -1.04                 | -1.06                 | <b>+1.74</b>          | <b>+3.68%</b>              |
| <b>Reacher / LeWM transformer</b> |                                 |                       |                       |                       |                            |
| Frozen LeWM                       | <b>48.4</b>                     | 35.9                  | 35.9                  | 17.7                  | 0.0591                     |
| Framewise calibration             | 42.7 $\pm$ 4.8                  | <b>42.7</b> $\pm$ 3.3 | <b>42.7</b> $\pm$ 3.3 | <b>21.2</b> $\pm$ 5.9 | 0.0092 $\pm$ 0.0001        |
| Shared context                    | 40.1 $\pm$ 3.9                  | 36.5 $\pm$ 3.9        | 36.5 $\pm$ 3.9        | 16.8 $\pm$ 3.3        | 0.0250 $\pm$ 0.0002        |
| Unpaired contexts                 | 38.0 $\pm$ 12.7                 | 34.9 $\pm$ 9.0        | 34.9 $\pm$ 9.0        | 20.8 $\pm$ 2.6        | 0.0075 $\pm$ 0.0002        |
| ShiftWM (ours)                    | 43.8 $\pm$ 4.7                  | 40.6 $\pm$ 3.1        | 40.6 $\pm$ 3.1        | 19.6 $\pm$ 1.3        | <b>0.0068</b> $\pm$ 0.0001 |
| Unaligned diagnostic              | 44.8 $\pm$ 4.5                  | 41.7 $\pm$ 3.9        | 41.7 $\pm$ 3.9        | 20.0 $\pm$ 2.3        | 0.0072 $\pm$ 0.0000        |
| Ours — Framewise / gain           | <b>+1.04</b>                    | -2.08                 | -2.08                 | -1.56                 | <b>+26.78%</b>             |

Table 39: **Historical context model: all core method–metric results.** Trained entries are three-seed means  $\pm$  sample SD; Frozen LeWM is one released checkpoint. Black bold marks the best point mean per environment and metric (including exact ties); the unaligned row remains a diagnostic, not an aligned baseline. Canonical is  $(v_0, p_0)$ ; held-out, eligible and MSE@5 use  $(v_2, p_2)$ ; extrapolation averages three registered conditions. Eligible success excludes tasks solved during common support. The gain rows compare ShiftWM (ours) with Framewise; the four success differences are percentage points, and the last column is relative MSE reduction in percent. **Bold green** means a favorable point difference, not significance. Dashes mean incomplete evidence. Paired planning uncertainty is unchanged in the existing contrast tables. These are Ours-1 results, not simulator evaluation of the newer spatial model.

| Environment | Population | Control           | Raw (%)          | Support | Eligible (%)     |
|-------------|------------|-------------------|------------------|---------|------------------|
| PushT       | Held-out   | Random actions    | 7.81 [5/64]      | 1/64    | 6.35 [4/63]      |
| PushT       | Held-out   | Privileged replay | 100.00 [64/64]   | 1/64    | 100.00 [63/63]   |
| PushT       | Extrap.    | Random actions    | 9.38 [18/192]    | 5/192   | 6.95 [13/187]    |
| PushT       | Extrap.    | Privileged replay | 100.00 [192/192] | 5/192   | 100.00 [187/187] |
| Reacher     | Held-out   | Random actions    | 4.69 [3/64]      | 0/64    | 4.69 [3/64]      |
| Reacher     | Held-out   | Privileged replay | 100.00 [64/64]   | 0/64    | 100.00 [64/64]   |
| Reacher     | Extrap.    | Random actions    | 4.69 [9/192]     | 0/192   | 4.69 [9/192]     |
| Reacher     | Extrap.    | Privileged replay | 100.00 [192/192] | 0/192   | 100.00 [192/192] |

Table 40: **Random-action and privileged-replay controls.** Raw and eligible columns show success percentages with successes/tasks in brackets; support counts are solved during paid history acquisition. Held-out is  $(v_2, p_2)$  (64 tasks); extrapolation pools the three prespecified conditions (192 tasks, 64 initial-state seeds). The task identities, goal index, support-success masks and recorded post-support diagnostics match the frozen-model reference. All policies share a 50-step native budget, including up to 10 support steps. Random actions use one fixed RNG rule (1701 plus initial-state seed), reused across appearances; these are not three training-seed estimates. Privileged replay uses recorded future actions and checks goal feasibility, not fair learned-policy inference. Neither control selects actions with model predictions, so CEM search metadata is unused. No independent-policy-seed uncertainty or significance claim is made.

### S.1 RANDOM ACTIONS AND PRIVILEGED REPLAY

These controls retain the same task identities, goal definition, paid support, and native interaction budget. Random actions sample uniformly within the native bounds using a single fixed RNG rule; they are not estimates over three trained checkpoints or multiple independent policy seeds. ShiftWM’s three-seed raw PushT success means are 2.60% on held-out composition and 6.77% on extrapolation, compared with random-action point estimates of 7.81% and 9.38%. On Reacher the corresponding ShiftWM means are 40.63% and 19.62%, compared with 4.69% for the random control on each population. These are descriptive comparisons; no significance or policy-seed uncertainty claim is inferred from their ordering.

Privileged replay uses the recorded future actions that generated each target. Its complete success verifies reachability under this replay protocol, not fair learned-policy performance or correctness of every learned planning operation. Neither control selects actions from world-model predictions, so CEM settings in their common evaluator metadata do not describe a search performed by those policies. Both raw and support-excluded eligible counts are retained in Table 40.

## T PAIRED PLANNING COMPARISONS

The paired reporter requires all three training seeds for both methods before releasing a contrast. It verifies the complete manifest-derived task population, source and protocol identities, support-success masks, and recorded initial distances across methods and training seeds. Initial-state clusters retain all their conditions and model outcomes during resampling. These intervals are conditional on the three observed training seeds; training-seed variability is reported separately. The main records do not contain support/goal pixel hashes or full simulator-state hashes, so those equalities are not retrospectively claimed. The reporting protocol is exploratory and does not select checkpoints or methods from test scores.

| Environment | Population | Comparator            | Raw $\Delta$ (pp)                | Eligible $\Delta$ (pp)           |
|-------------|------------|-----------------------|----------------------------------|----------------------------------|
| PushT       | Held-out   | Framewise calibration | -1.0 $\pm$ 0.9                   | -1.1 $\pm$ 0.9                   |
|             |            |                       | [-3.6, +1.6]                     | [-3.7, +1.6]                     |
|             |            |                       | <b>+0.5 <math>\pm</math> 0.9</b> | <b>+0.5 <math>\pm</math> 0.9</b> |
| PushT       | Held-out   | Shared context        | [-1.0, +2.1]                     | [-1.1, +2.6]                     |
|             |            |                       | +0.0 $\pm$ 0.0                   | +0.0 $\pm$ 0.0                   |
|             |            | Unpaired contexts     | [-2.1, +2.1]                     | [-2.1, +2.1]                     |
| PushT       | Held-out   |                       | <b>+1.7 <math>\pm</math> 0.3</b> | <b>+1.8 <math>\pm</math> 0.3</b> |
| PushT       | Extrap.    | Framewise calibration | [+0.0, +3.6]                     | [+0.0, +3.8]                     |
|             |            |                       | -0.9 $\pm$ 1.1                   | -0.9 $\pm$ 1.1                   |
|             |            | Shared context        | [-3.0, +1.2]                     | [-3.0, +1.1]                     |
| PushT       | Extrap.    |                       | <b>+1.6 <math>\pm</math> 0.9</b> | <b>+1.6 <math>\pm</math> 0.9</b> |
|             |            | Unpaired contexts     | [-0.5, +3.6]                     | [-0.5, +3.8]                     |
|             |            |                       | -2.1 $\pm$ 0.9                   | -2.1 $\pm$ 0.9                   |
| Reacher     | Held-out   | Framewise calibration | [-11.5, +7.3]                    | [-11.5, +7.3]                    |
|             |            |                       | <b>+4.2 <math>\pm</math> 5.9</b> | <b>+4.2 <math>\pm</math> 5.9</b> |
|             |            | Shared context        | [-4.2, +12.5]                    | [-4.2, +12.5]                    |
| Reacher     | Held-out   |                       | <b>+5.7 <math>\pm</math> 6.5</b> | <b>+5.7 <math>\pm</math> 6.5</b> |
|             |            | Unpaired contexts     | [-2.6, +14.1]                    | [-2.6, +14.1]                    |
|             |            |                       | -1.6 $\pm$ 5.0                   | -1.6 $\pm$ 5.0                   |
| Reacher     | Extrap.    | Framewise calibration | [-6.2, +3.1]                     | [-6.2, +3.1]                     |
|             |            |                       | <b>+2.8 <math>\pm</math> 4.4</b> | <b>+2.8 <math>\pm</math> 4.4</b> |
|             |            | Shared context        | [-1.9, +7.6]                     | [-1.9, +7.6]                     |
| Reacher     | Extrap.    |                       | -1.2 $\pm$ 3.3                   | -1.2 $\pm$ 3.3                   |
|             |            | Unpaired contexts     | [-5.7, +3.3]                     | [-5.7, +3.3]                     |

Table 41: Exploratory paired planning contrasts: ShiftWM (ours) minus comparator, in percentage points. Entries are mean  $\pm$  sample SD across the three fixed training seeds, with conditional 95% initial-state-cluster bootstrap intervals on the second line (20,000 draws). Green bold means indicate positive differences against the named comparator, not statistical significance. Each resampled cluster retains all conditions and all three training seeds. Eligible success excludes tasks solved during paid support acquisition. Dashes require further completed sources; no partial-seed contrasts are shown. Intervals do not capture population-level training uncertainty, are not multiplicity-adjusted, and a degenerate interval does not demonstrate equivalence.

### Does better prediction translate into better planning? ShiftWM (ours) minus each matched control



Figure 19: **Paired planning differences.** Points show mean paired raw success differences across three training seeds; positive values favor ShiftWM (ours). Bars are 95% initial-state-cluster bootstrap intervals conditional on those trained models, not training-seed standard deviations or uncertainty over the training population. Pending comparisons have no point estimate. These exploratory intervals are not multiplicity-adjusted.

## U DEVELOPMENT REVISION WITH FIXED VISUAL CALIBRATION

The completed goal-only intervention motivates a separate, development-only question: whether a trained dynamics residual can improve an otherwise frozen framewise world model. This revision freezes the completed framewise visual calibrator, goal pathway, action encoder, and predictor, including normalization statistics and dropout mode. It adds a transition-context network and a zero-initialized action-embedding FiLM residual. Before optimization, the revision reproduces the donor’s predictions; only the new 218,272 parameters receive gradients. It reuses the original training/validation data, query indexing, action normalization, and 30-epoch optimizer schedule, selecting by validation prediction error. No main-test outcomes select this revision.

These extra optimization steps make this an exploratory extension, not a matched-training-budget comparison. An improvement over the frozen donor would not by itself establish that temporal conditioning is necessary: a separately trained context-free residual remains an important control. The table keeps uncompleted training or development planning missing and never merges the revision with the original three-seed study.

| Environment | Model                      | Raw   | Support | Eligible | Wins/losses |
|-------------|----------------------------|-------|---------|----------|-------------|
| PushT       | Framewise calibration      | 2/32  | 1/32    | 1/31     | —           |
| PushT       | + Dynamics residual (ours) | 2/32  | 1/32    | 1/31     | 1/1         |
| Reacher     | Framewise calibration      | 13/32 | 2/32    | 11/30    | —           |
| Reacher     | + Dynamics residual (ours) | 10/32 | 2/32    | 8/30     | 4/7         |

Table 42: **Post-hoc dynamics-residual development experiment.** Starting from the completed seed-0 framewise model, the visual calibrator and predictor stay frozen while a new dynamics-context/action-embedding residual receives 30 additional training epochs. Both policies use the same 32 development tasks and common support, with 300 CEM candidates, 30 iterations and 30 elites. Raw success includes support-only success; eligible success excludes it. Wins/losses count eligible tasks solved only by the revision/control. Dashes for the control denote no self-comparison; pending cells mean incomplete evidence. This uses extra optimization, not matched training budgets, and does not establish context necessity, physical identification, or a main-test result.

## V DEVELOPMENT STUDY OF ROLLOUT OBJECTIVE AND CONTEXT DEPENDENCE

The completed dynamics-residual development experiment does not improve its frozen Framewise donor: eligible success is 1/31 for both models on PushT and 8/30 versus 11/30 on Reacher. We therefore record a new, post-hoc development protocol rather than promote that revision to the main test. All eight arms have completed training and development planning in separate environment campaigns. A  $2 \times 2$  design crosses teacher-forced versus recursive prediction with inferred versus constant-input dynamics context. Each of the four variants per environment trains for 30 epochs at seed 0, freezing the same completed Framewise calibrator, goal pathway, action encoder and predictor. Both objectives supervise target frames 3–7 (indices 3 : 8), including the first prediction after the three-frame support. Teacher forcing uses available observed history for each target; recursive training feeds back predicted latents after support. Context uses support only. The constant-input control retains the identical GRU/FiLM architecture but receives all-zero support and executed-action inputs: its learned response can be nonzero, and matching nominal parameter counts does not establish equal effective capacity. All variants reuse the existing train/validation data and optimizer schedule; validation-selected checkpoints minimize the same FP32 recursive MSE averaged over all five query frames. Development planning uses the existing 32 matched tasks per environment, 300 CEM candidates, 30 iterations, 30 elites and the fixed 50-native-step budget including support. All eight prespecified arms are retained, and only validation error selects checkpoints within this follow-up protocol. Table 43 reports source-validated status and distinguishes incomplete training, trained checkpoints and completed planning; missing measurements are never substituted with zeros or partial planning outcomes. The completed teacher-forced arms give mixed evidence about context dependence. Inferred context ties the frozen donor on PushT (1/31 eligible successes) and Reacher (11/30). Constant input scores 0/31 on PushT and 15/30 on Reacher, compared with donor scores of

1/31 and 11/30, respectively. The latter is a favorable development point estimate for the constant-input control, not evidence that inferred context is necessary. The completed recursive/inferred arms tie the donor on PushT (1/31) and score 13/30 versus 11/30 on Reacher (a +6.67 percentage-point development difference). Recursive constant context scores 0/31 on PushT and 13/30 on Reacher. Thus, inferred context exceeds its constant-input control on PushT but ties the donor; on Reacher, recursive inferred and constant contexts tie. Neither objective satisfies the registered screening requirement to exceed both its constant-input control and donor in both environments. Lower recursive validation error does not establish better control, and these results do not establish a benefit from episode-dependent dynamics inference. Repeated training seeds and an untouched evaluation remain necessary to assess any subsequent method. A revised context method is not promoted without evidence of learning beyond its matched constant-input control; multiple training seeds and a fresh final holdout unused during revision development are required before a broader performance claim. These experiments remain separate from the original three-seed campaign and its negative outcomes.

| Environment | Objective      | Context  | Status   | Val. MSE | Eligible | $\Delta$ (pp) |
|-------------|----------------|----------|----------|----------|----------|---------------|
| PushT       | Teacher-forced | Inferred | Complete | 0.28759  | 1/31     | +0.00         |
| PushT       | Teacher-forced | Constant | Complete | 0.28791  | 0/31     | -3.23         |
| PushT       | Recursive      | Inferred | Complete | 0.26388  | 1/31     | +0.00         |
| PushT       | Recursive      | Constant | Complete | 0.26876  | 0/31     | -3.23         |
| Reacher     | Teacher-forced | Inferred | Complete | 0.00566  | 11/30    | +0.00         |
| Reacher     | Teacher-forced | Constant | Complete | 0.00565  | 15/30    | <b>+13.33</b> |
| Reacher     | Recursive      | Inferred | Complete | 0.00521  | 13/30    | <b>+6.67</b>  |
| Reacher     | Recursive      | Constant | Complete | 0.00525  | 13/30    | <b>+6.67</b>  |

Table 43: **Development objective/context study: protocol and completion status.** Eight seed-zero arms train for 30 extra epochs with a frozen Framewise donor. Planned means completed training is unavailable; trained means validated training is complete but planning is incomplete; complete requires all 32 matched development tasks. Validation MSE is the completed-history minimum of the common FP32 recursive criterion over targets 3–7, not a test score. Eligible success excludes support-only successes;  $\Delta$  compares the same eligible tasks against the frozen Framewise donor. Bold green marks strictly positive observed differences, not statistical significance. Missing values are unavailable, never zero. Constant-input contexts can learn nonzero responses; nominal parameter matching does not establish equal effective capacity. Both objectives use the first target after support. Planning uses 300 CEM candidates, 30 iterations, 30 elites and the fixed native-action budget. This post-hoc development study does not establish a main-test, multi-seed, efficiency or context-necessity claim.

## W REGISTERED DOMAIN AND PREDICTOR EXTENSIONS

The original forecasting gains, mixed planning comparisons, and failed rollout-context screen remain unchanged. A separately registered 36-run development campaign has completed training, forecasting and planning on two additional simulations and a second predictor family. A subsequent six-run observation-gain ablation is also complete. These adapted tasks are not reproductions of published drone or surgical leaderboards. Table 44 distinguishes completed development evaluation from final testing, which remains unrun. All positive and negative outcomes below remain separate from the original test results.

| Stage                              | Drone             | Tissue manipulation      |
|------------------------------------|-------------------|--------------------------|
| Recorded trajectories              | 384 collected     | 384 collected            |
| Full collection audit              | Passed            | Passed                   |
| Reachable-goal protocol            | Audited           | 112 replay checks passed |
| Frozen visual feature cache        | Complete          | Complete                 |
| Full 30-epoch training             | 18/18 complete    | 18/18 complete           |
| Development forecasting            | 18/18 complete    | 18/18 complete           |
| Development planning               | 18/18 complete    | 18/18 complete           |
| Gain ablation: train/forecast/plan | 6/6 in each stage | Not registered           |
| Final model test evaluation        | Pending           | Pending                  |

Table 44: **Completed registered extension campaigns.** Each original domain has two predictor families, three methods and three training seeds. The subsequent drone ablation has two methods and three seeds. All models completed 30 epochs and both development evaluations; pending final tests are unavailable, not zero performance.

**Data and physical tasks.** Each domain records 384 trajectories of 200 native commands, partitioned into 192 training, 48 validation, 48 development and 96 test trajectories. Initial-state seed families are disjoint between splits; within a split, the same seeds are recorded across response gains 1, 0.75 and 1.25. Five successive two-dimensional commands form one model action block. Model inputs contain RGB observations and commanded actions. Simulator state is retained separately for collection verification and scoring. Action normalization uses only the training split. Photometric transforms and the seven training/validation appearance–gain combinations follow the original loader; development uses warm appearance with gain 0.75, while cool appearance with gain 1.25 is withheld from training.

The drone task adapts `gym-pybullet-drones` (Panerati et al., 2021), pinned at revision `7ebad1ecabd2`. A CF2X vehicle receives planar velocity requests through the upstream PID controller at fixed requested altitude, observed by a fixed external RGB camera. The intervention scales commanded response; it is not a wind or mass change. Recorded settled endpoints define goals. Success requires XY distance below 4 cm, speed below 6 cm/s, altitude error below 5 cm, and no crash or workspace escape. The complete collection audit checked all 384 trajectories, payload hashes and 77,184 native states.

The deformable-tissue task adapts LapGym’s SOFA tissue manipulation environment (Scheikl et al., 2023), pinned at revision `85bf7e05dd08`. Two-dimensional gripper commands are scaled before native clipping. Forecast collection and planning-goal verification are separate: planning goals require an independent recorded-action replay with the desired marker moved to the recorded endpoint, preserving the physical trajectory. Native XZ tissue-point distance at most 2 mm defines success. All 384 collection episodes passed the full data audit; independent replays verified all 112 eligible development/test goal references, including physical-state equality and RGB equality outside the relocated target marker. Finite workspace-clipped collection transitions are retained; planning eligibility additionally excludes unstable or invalid reference trajectories before model evaluation. This is a constrained reachable-goal distribution, not clinical validation or a claim of autonomous surgical safety.

**Predictor families and matched controls.** Both families reuse the frozen released PushT LeWM visual encoder and its canonical target coordinates; neither uses drone or surgical visual pretraining. The first retains the released transformer initialization. The second replaces its predictor with a two-layer GRU of hidden size 256, using the same projected state/action interface. The GRU is an in-house architecture control, not a Dreamer reproduction. Parameter counts and initialization differ across families; comparisons are primarily within each family. The three methods are *Framewise calibration*, *ShiftWM (ours)*, and *Constant dynamics context*. The last retains episode-dependent observation correction but supplies zero inputs to the dynamics-context network. Its learnable biases permit a nonzero common response; equal nominal parameter count does not imply equal effective capacity. Seeds 0, 1 and 2 give 18 runs per domain.

Every run trains for 30 epochs with AdamW, learning rate  $10^{-4}$ , cosine minimum  $10^{-6}$ , weight decay 0.01, gradient clipping 1, batch size 128 and stride 4. Eight-frame windows provide three observed support frames and five recursive query targets, including the first unobserved target. Training

uses BF16 where supported; selection uses the minimum full-validation recursive MSE in float32 (autocast disabled; CUDA TF32 matrix multiplication remains allowed for every arm). Alignment and context consistency weights match across the applicable methods. Source, data, cache, normalization and checkpoint identities are pinned; optimizer and RNG state support continuation. Training completion alone is not a planning result or a validated release.

**Development planning and decision rule.** Before model outcomes, select the first eight eligible development trajectories in seed order at gain 0.75, or report a shortfall if fewer are available. Initial-goal separation must be at least 8 cm for the drone and 4 mm for tissue manipulation; initially solved tasks are excluded and every exclusion is retained. Policy-induced invalid actions count as failures. All methods share goal images, ten recorded support commands, planner random numbers and a 200-native-command total budget including support. CEM uses horizon five blocks, 128 candidates, five iterations and 16 elites; it replans after at most five executed commands and retains the unused plan as its warm start. Search uses standardized action coordinates, with clipping after conversion to native commands. Report support-only successes separately and log partial final blocks exactly.

Random actions use the same interaction budget; privileged replay of recorded actions checks goal reachability. Replay is not a deployable baseline. We report success counts and rates, paired task and seed uncertainty, native interactions, physical errors, failures and latency. Promotion requires better closed-loop success than both within-family controls with consistent evidence across seeds and domains. Final test evaluation is separate from development, and any changed hypothesis or protocol receives a new run namespace. This campaign cannot turn the original null planning findings into positive evidence by selectively reporting a favorable domain or model.

## W.1 COMPLETED DEVELOPMENT COMPARISONS

All 36 conditions have completed training, forecasting and planning. Table 45 reports every method and training seed for both domains, with the six subsequent wide-gain runs in a separate group. No task succeeds during the ten-command common support. On the drone, transformer ShiftWM achieves 20.83% success versus 16.67% for either control; with the GRU it achieves 16.67%, versus 25.00% for Framewise and 29.17% for Constant dynamics. On tissue manipulation, transformer ShiftWM achieves 4.17% versus 8.33% for either control, while GRU ShiftWM achieves 12.50% versus 8.33% for either control. These opposite signs across families and domains do not establish a general benefit of inferred dynamics context.

Table 46 retains the matched task structure when quantifying uncertainty: it jointly resamples training seeds and physical task seeds in 5,000 crossed bootstrap draws. No original interval has a strictly positive lower bound. Forecast MSE and planning success need not agree: drone GRU ShiftWM has lower horizon-five error than either control but lower closed-loop success. Final test results are not used here, and these development results do not satisfy the promotion rule above.

The fixed random-action control succeeds on 2/8 drone and 1/8 tissue tasks; privileged recorded-action replay succeeds on 8/8 in each domain. Random uses base seed 101 plus each task seed, with no independently trained models; replay is a reachability check. Original drone ShiftWM in both predictor families is below the random 25% point estimate. Neither this drone task nor the tissue simulation establishes useful real-world control or clinical performance.

| Method                                                      | Success $\uparrow$ |              | MSE ( $10^{-3}$ ) $\downarrow$ |              |              | Gain vs ref.  |                 |  |
|-------------------------------------------------------------|--------------------|--------------|--------------------------------|--------------|--------------|---------------|-----------------|--|
|                                                             | S0/1/2             | %            | h1                             | h3           | h5           | pp $\uparrow$ | h5 % $\uparrow$ |  |
| <i>Drone / Transformer; ref.: Framework</i>                 |                    |              |                                |              |              |               |                 |  |
| Framework                                                   | 1,2,1              | 16.67        | <b>6.403</b>                   | <b>5.975</b> | <b>5.650</b> | ref.          | ref.            |  |
| Constant dynamics                                           | 1,1,2              | 16.67        | 6.479                          | 6.049        | 5.718        | +0.00         | -1.21           |  |
| ShiftWM (ours)                                              | 1,2,2              | <b>20.83</b> | 6.478                          | 6.047        | 5.711        | <b>+4.17</b>  | -1.08           |  |
| <i>Drone / GRU; ref.: Framework</i>                         |                    |              |                                |              |              |               |                 |  |
| Framework                                                   | 1,3,2              | 25.00        | <b>6.509</b>                   | 6.996        | 6.578        | ref.          | ref.            |  |
| Constant dynamics                                           | 1,4,2              | <b>29.17</b> | 6.571                          | 7.027        | 6.591        | <b>+4.17</b>  | -0.20           |  |
| ShiftWM (ours)                                              | 1,2,1              | 16.67        | 6.597                          | <b>6.975</b> | <b>6.503</b> | -8.33         | <b>+1.13</b>    |  |
| <i>Tissue manipulation / Transformer; ref.: Framework</i>   |                    |              |                                |              |              |               |                 |  |
| Framework                                                   | 0,1,1              | <b>8.33</b>  | <b>2.823</b>                   | 3.540        | 3.588        | ref.          | ref.            |  |
| Constant dynamics                                           | 1,1,0              | <b>8.33</b>  | 2.851                          | <b>3.529</b> | <b>3.563</b> | +0.00         | <b>+0.69</b>    |  |
| ShiftWM (ours)                                              | 1,0,0              | 4.17         | 2.853                          | 3.542        | 3.573        | -4.17         | <b>+0.41</b>    |  |
| <i>Tissue manipulation / GRU; ref.: Framework</i>           |                    |              |                                |              |              |               |                 |  |
| Framework                                                   | 0,1,1              | 8.33         | <b>3.696</b>                   | <b>4.462</b> | 4.405        | ref.          | ref.            |  |
| Constant dynamics                                           | 0,1,1              | 8.33         | 3.736                          | 4.488        | 4.417        | +0.00         | -0.26           |  |
| ShiftWM (ours)                                              | 0,1,2              | <b>12.50</b> | 3.726                          | 4.470        | <b>4.389</b> | <b>+4.17</b>  | <b>+0.38</b>    |  |
| <i>Drone / Transformer / wide gain; ref.: wide Constant</i> |                    |              |                                |              |              |               |                 |  |
| Constant dynamics                                           | 0,2,2              | 16.67        | 6.474                          | 6.053        | 5.724        | ref.          | ref.            |  |
| ShiftWM (ours)                                              | 1,3,3              | <b>29.17</b> | <b>6.469</b>                   | <b>6.046</b> | <b>5.713</b> | <b>+12.50</b> | <b>+0.20</b>    |  |

Table 45: **Historical context model: complete extension scorecard.** The original 36 models and six wide-gain follow-ups are grouped by domain, backbone and recipe. S0/1/2 gives success counts out of eight physical tasks per training seed; 24 task–seed instances are not 24 independent tasks. MSE averages three seeds on the same 16 development trajectories and 144 windows per domain. Black bold marks each group’s best point mean. Named-reference gains are success percentage points and relative h5 error reductions; green bold marks favorable point differences, not significance. All 42 runs completed 30 epochs and both evaluations. Paired uncertainty is retained separately. Final model tests and comparable physical-error aggregates remain pending. These are historical-model development results.

| Domain | Predictor   | Comparator        | $\Delta$ (pp) | Paired 95% interval (pp) |
|--------|-------------|-------------------|---------------|--------------------------|
| Drone  | Transformer | Framework         | <b>+4.17</b>  | [−25.00, +33.44]         |
| Drone  | Transformer | Constant dynamics | <b>+4.17</b>  | [−25.00, +33.33]         |
| Drone  | GRU         | Framework         | -8.33         | [−29.17, +12.50]         |
| Drone  | GRU         | Constant dynamics | -12.50        | [−37.50, +12.50]         |
| Tissue | Transformer | Framework         | -4.17         | [−25.00, +12.50]         |
| Tissue | Transformer | Constant dynamics | -4.17         | [−20.83, +0.00]          |
| Tissue | GRU         | Framework         | <b>+4.17</b>  | [+0.00, +25.00]          |
| Tissue | GRU         | Constant dynamics | <b>+4.17</b>  | [+0.00, +25.00]          |

Table 46: **ShiftWM (ours) minus matched controls in planning success.** Differences are percentage points. **Bold green** marks positive point differences, not statistical significance. Intervals use paired crossed resampling of three training and eight physical task seeds; they are unadjusted exploratory 95% intervals. All intervals include or touch zero, and all negative comparisons remain visible.

## W.2 GOAL GEOMETRY AND CONTROLLED ACTION-RANKING DIAGNOSTICS

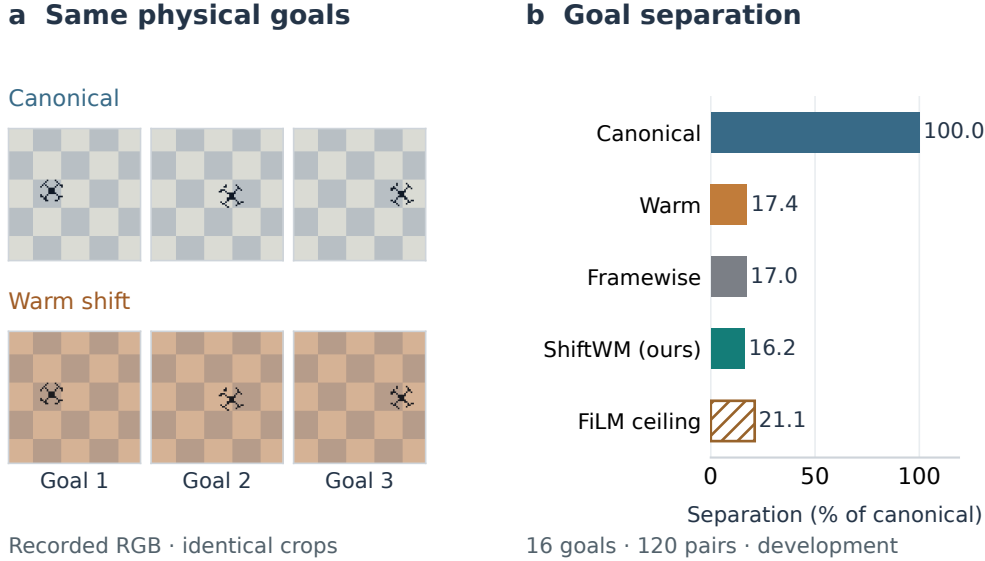
The following diagnostics use only training/development evidence and were motivated by the unsuccessful planning comparisons. They distinguish a representational limitation from an identified cause of failure.

**A fixed-context capacity bound.** Across all sixteen gain-0.75 development endpoint goals (120 pairs), the warm appearance transform reduces mean pairwise latent MSE to 17.4% of the canonical

value. In Figure 20, the transformer seed-0 Framewise and ShiftWM corrections retain 17.0% and 16.2%, respectively. For each observation context, all goals receive the same correction; measurements are then averaged over sixteen contexts. If  $G(u, c) = g(c) \odot u + b(c)$  and each coordinate of  $g(c)$  lies in  $[0.9, 1.1]$ , its shared translation cancels and

$$0.81\|u - v\|_2^2 \leq \|G(u, c) - G(v, c)\|_2^2 \leq 1.21\|u - v\|_2^2. \quad (17)$$

Thus the existing observation FiLM can expand this warm-goal mean separation to at most 21.1% of its canonical value in the fixed-context comparison. This is a capacity bound on pairwise separation, not an achieved result, an uncertainty interval, or a proof that larger distances improve control. Framewise calibration lacks this FiLM bound but also fails to recover canonical separation in this diagnostic. Canonical features themselves only imperfectly order physical distance, which is only one component of the drone success criterion.



**Figure 20: Development goal geometry under the appearance shift.** Left: three recorded drone goals selected by minimum, median and maximum X position, shown with identical central  $80 \times 80$  crops and the registered warm transform. Right: mean pairwise latent MSE for all sixteen goals, normalized by canonical separation; correction uses fixed support contexts and transformer seed 0. The hatched bar is the analytic  $1.21\times$  expansion ceiling for original observation FiLM applied to warm features. It is not a new-method measurement or a causal explanation of planning failure.

**Four-state counterfactual action probe.** Six seed-0 models score the same fixed library of 32 candidate sequences from each of four development support states before any candidate is executed. Each of the 128 branches independently resets and exactly replays the ten-command support, then executes up to 25 future commands. Two branches reach a workspace boundary before the full horizon; all branches are retained, while equal-horizon, safe branches define the within-task rank correlations and regret. All 24 model-selected winners are safe. This probe uses actual alternative action executions from a common state, not merely same-seed trajectories collected under different response gains: the latter use state-dependent feedback and do not have identical action histories.

Realized latent cost compares the canonical encoding of each executed future image with that model’s adapted goal; these future images are used only for the post-execution audit. For transformer ShiftWM, mean Spearman correlation between predicted and realized latent goal costs is 0.422; for GRU ShiftWM it is 0.278. Correlations are computed within each task and then averaged over the four tasks. Realized latent-cost versus physical-distance correlations are approximately 0.675, compared with 0.814 using frozen canonical feature-to-goal costs. The transformer ShiftWM

and Constant dynamics models choose the same candidate on all four tasks, with mean terminal distance 61.7 mm and regret 40.9 mm relative to the best safe candidate. Transformer Framework gives 66.4 mm and 45.6 mm; all three GRU variants choose the same candidates, giving 157.5 mm and 136.6 mm. The 177.4 mm uniform mean over the candidate library is a descriptive reference, not the random closed-loop controller. These  $n = 4$  observations suggest both imperfect action-conditioned ranking and imperfect goal geometry; they neither isolate a unique mechanism nor establish performance on a larger task population. In particular, identical selected actions for inferred and constant contexts do not demonstrate a dynamics-inference benefit.

**Completed, separately registered capacity ablation.** The six-run transformer study changes only the observation gain to  $g(s) = \exp\{\log(4) \tanh[0.1s/\log(4)]\}$ , comparing inferred and constant dynamics context at seeds 0, 1 and 2. The attainable gain range becomes  $[0.25, 4]$ , while identity initialization, derivative 0.1 at zero, parameter shapes, translation, action modulation, losses and the complete 30-epoch training schedule remain matched. Higher-order curvature also changes, so optimization trajectories need not match. This is a capacity ablation, not a claim of algorithmic novelty. Its versioned checkpoints and output namespace are separate from the original study.

All six runs completed 30 epochs and both development evaluations (Table 47). Revised ShiftWM reaches 7/24 task–seed instances, versus 4/24 for the revised Constant dynamics control: +12.50 percentage points, with a paired exploratory interval  $[0, 37.50]$ . Each training seed gains one of eight successes, with three ShiftWM-only successes and no control-only success across the matched instances. Relative to original ShiftWM (5/24), there are three new successes and one lost success. The difference is +8.33 points, with interval  $[-12.50, 29.17]$ . The revised Constant dynamics control retains 4/24 in aggregate but changes which tasks succeed. The revised ShiftWM rate, 29.17%, exceeds the single random-policy reference of 25% only descriptively.

These are subsequent comparisons on the same development tasks. The forecast MSE changes little, and the fixed 16-goal/120-pair geometry measurement has not yet been repeated for the revised checkpoints. Consequently, the observed success increases do not demonstrate restored goal geometry, improved action ranking, or independent generalization. They do not justify promoting the method to final-test evaluation.

| Paired contrast                               | $\Delta$ (pp) | 95% interval (pp)  |
|-----------------------------------------------|---------------|--------------------|
| Wide ShiftWM (ours) — wide constant           | <b>+12.50</b> | $[+0.00, +37.50]$  |
| Wide ShiftWM (ours) — original ShiftWM (ours) | <b>+8.33</b>  | $[-12.50, +29.17]$ |
| Wide constant — original constant             | +0.00         | $[-25.00, +33.33]$ |

Table 47: **Completed observation-gain ablation on drone development.** Original and revised transformer conditions share data, optimization, validation selection, eight physical tasks, support and planner budget. The six revised models use seeds 0, 1 and 2. Absolute scores are consolidated in Table 45; this table retains paired crossed-bootstrap success differences. All intervals include or touch zero. **Bold green** marks positive point estimates only. This follow-up ablation is exploratory, not an independent test of the geometry hypothesis.

### W.3 SEPARATE OFFICIAL ADAJEPa REPRODUCTION

We additionally ran the released AdaJEPa implementation (Wang et al., 2026c), pinned at revision 51d8665b7978, with its official PushT checkpoint and evaluation goals. All four 50-episode evaluations completed (Table 48). The model observes RGB plus agent proprioception; the released goals, pretrained representation and planner budget differ from our RGB-only studies. These results therefore remain a separate reproduction, without a direct performance ranking against ShiftWM.

Official CEM uses horizon 25, 200 candidates, 30 elites and ten optimization iterations, with at most 20 replans and five grouped actions executed per replan (frameskip five). All stages use evaluation seed 100. Adaptive success exceeds frozen success by 24 percentage points on clean observations and 20 under blur. Aggregate logs and one evaluation seed do not support a paired task-level uncertainty interval. This is evidence that the released adaptive implementation runs locally under its own protocol, not a state-of-the-art or matched-baseline claim.

The checkpoint, goals, official configuration and scientific command are pinned for every stage. The first clean-frozen run began before launcher/runtime hashes were added to the record; a supplemental GPU observation documents its hardware. Later stages include those metadata fields, but current hashes cannot retroactively establish their exact values at the first stage’s launch.

| Observation | Official frozen | Official adaptive | $\Delta$ (pp) |
|-------------|-----------------|-------------------|---------------|
| Clean       | 34/50 (68%)     | 46/50 (92%)       | <b>+24.00</b> |
| Blur        | 29/50 (58%)     | 39/50 (78%)       | <b>+20.00</b> |

Table 48: **Official AdaJEPa reproduction under its separate protocol.** Each entry uses 50 released evaluation goals and seed 100, with RGB plus agent proprioception. Differences compare adaptive and frozen execution within this official protocol. **Bold green** marks positive aggregate point differences; no paired interval or significance claim is made. These inputs and budgets are not matched to ShiftWM.

## X ARTIFACT PROVENANCE

The earlier TTA draft is retained under `paper/archive_tta`. This manuscript uses the unmodified official ICLR 2027 style files. Architecture and split diagrams have editable TikZ sources, a figure brief, and a review record. Numerical plots are populated only from recorded evidence. Every plotted value and uncertainty interval must link to a run record; unavailable values remain missing.

### X.1 REUSABLE MODELS AND SCOPE

The release target is a full weights-only model, preprocessing and action normalization, history/reset rules, base revision and hash, and a planning example for each validated environment family. Adapter-only weights require the corresponding base hash. Current packages preserve every loaded tensor; the simulation release stores identical tensors once across models. Active-path counts do not measure archive size or runtime memory. Portable local frozen exports and validation-selected checkpoints from the completed 30-epoch runs have been load-verified. The existing local demo uses recorded held-out simulator clips and measured latent errors. The public real-DROID release contains twelve validation-selected predictors, the shared frozen encoder, and twelve training-fitted calibration wrappers, with model cards, source, licenses and checksums. All base models and calibrated variants pass exact offline prediction parity after relocation. The separate generalization development release adds all 36 predictors from Appendix F, including every comparison arm, with the shared encoder, source and checksums. Each passes an isolated offline CPU reload check. These packages remain labeled as validation-selected development models; they do not replace the original test checkpoints. A separate simulator development release provides 42 predictors: the 36 original extension models and six drone geometry revisions. All completed 30 epochs; their selected states reproduce source predictions exactly after offline relocation on both cached-feature and image inputs. Together, these five releases provide 117 trained predictors, including all mixed outcomes. The fourth release contains the twelve ten-step-trained predictors from Appendix H; all selected epoch one after full 30-epoch runs and passed exact relocated CPU prediction parity. The fifth release contains all fifteen spatial predictors from Appendix B. Each completed 30 epochs and passed exact relocated CPU prediction parity; all public archive payload hashes were independently verified against its release manifest. These models forecast visual features; evaluation status remains separate from artifact availability.

### X.2 VISUAL AND DIAGNOSTIC INTERPRETATION

The original composition illustration’s generated scene is an illustration, not simulator output; the evaluated appearance changes are pointwise RGB transforms, not viewpoint changes. Method feature strips depict correction, not reconstructed images or guaranteed canonical recovery. Candidate branches are schematic latents, not predicted videos or physical trajectories. Split response glyphs are not measured trajectories or ranked physical responses. Numerical plots use source-linked run records, not generated artwork.

The nominal simulator, collection policy, and interaction protocol differ from the released-evaluator check, so their performance gap cannot be attributed solely to appearance/dynamics shifts. The goal-only hybrid was chosen after inspecting development failures; its task-bootstrap intervals are exploratory summaries conditional on that selection. Any further revision motivated by these diagnostics remains separate from the fixed primary campaign and cannot be tuned on its test outcomes.

## Y TRAINING CAMPAIGN SNAPSHOT

| Environment | Mode                             | Seed | Epochs | State    |
|-------------|----------------------------------|------|--------|----------|
| reacher     | ShiftWM (ours)                   | 0    | 30/30  | complete |
| reacher     | Unaligned predictor (diagnostic) | 0    | 30/30  | complete |
| reacher     | Shared context                   | 0    | 30/30  | complete |
| pusht       | ShiftWM (ours)                   | 0    | 30/30  | complete |
| pusht       | Unaligned predictor (diagnostic) | 0    | 30/30  | complete |
| pusht       | Shared context                   | 0    | 30/30  | complete |
| reacher     | ShiftWM (ours)                   | 1    | 30/30  | complete |
| reacher     | Unaligned predictor (diagnostic) | 1    | 30/30  | complete |
| reacher     | Shared context                   | 1    | 30/30  | complete |
| pusht       | ShiftWM (ours)                   | 1    | 30/30  | complete |
| pusht       | Unaligned predictor (diagnostic) | 1    | 30/30  | complete |
| pusht       | Shared context                   | 1    | 30/30  | complete |
| reacher     | ShiftWM (ours)                   | 2    | 30/30  | complete |
| reacher     | Unaligned predictor (diagnostic) | 2    | 30/30  | complete |
| reacher     | Shared context                   | 2    | 30/30  | complete |
| pusht       | ShiftWM (ours)                   | 2    | 30/30  | complete |
| pusht       | Unaligned predictor (diagnostic) | 2    | 30/30  | complete |
| pusht       | Shared context                   | 2    | 30/30  | complete |
| reacher     | Framewise calibration            | 0    | 30/30  | complete |
| reacher     | Unpaired contexts                | 0    | 30/30  | complete |
| pusht       | Framewise calibration            | 0    | 30/30  | complete |
| pusht       | Unpaired contexts                | 0    | 30/30  | complete |
| reacher     | Framewise calibration            | 1    | 30/30  | complete |
| reacher     | Unpaired contexts                | 1    | 30/30  | complete |
| pusht       | Framewise calibration            | 1    | 30/30  | complete |
| pusht       | Unpaired contexts                | 1    | 30/30  | complete |
| reacher     | Framewise calibration            | 2    | 30/30  | complete |
| reacher     | Unpaired contexts                | 2    | 30/30  | complete |
| pusht       | Framewise calibration            | 2    | 30/30  | complete |
| pusht       | Unpaired contexts                | 2    | 30/30  | complete |